



Review article

Quantum walk and its application domains: A systematic review

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ABSTRACT

Quantum random walk is the quantum counterpart of a classical random walk. The classical random walk concept has long been used as a computational framework for designing classical algorithms for complex problems. Quantum analogues of random walk provide speed-up in computational power for various algorithms such as element distinctness, spatial search, graph connectivity, etc. Quantum walks have emerged to be a universal computational model over the last decade. Quantum walk formulations applied in graph theory have shown quadratic and polynomial time in graph traversal as opposed to the exponential time taken by classical algorithms. Quantum walk models have also found use in designing quantum computers. Inspired by these facts, this article presents a substantial systematic literature review and analysis of various quantum walk formulations and their strengths and limitations w.r.t. application domains used in literature up-to-date by researchers in various fields. The analysis provided in this article may help upcoming researchers to gain new insights towards the application of quantum walk formulation in varied domains. Various performance metrics, physical implementation set-ups, coin operators, and simulators used to analyze classical and quantum walk dynamics on graphs have been described. Finally, the article discusses existing open problems and notable future directions related to quantum walk application for potential researchers.

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1. Introduction and motivation

Quantum physics has revolutionized information processing in all spheres of science, particularly in computer science and engineering. These subject areas form the backbone of our technology era and have transformed all possible aspects of modern society. In these fields, leading edge research is mostly about innovation in hardware and software components. Devising new technology,

new models of computation, faster and more efficient algorithms have become a major concern with the ever-growing hunger for optimal utilization of computational power.

Inserting quantum mechanics concepts in classical models of computation pave the way towards quantization of existing models and algorithms as depicted in Fig. 1. The resultant quantum algorithms such as Shor's algorithm in 1994 [1] for factorization provided exponential speed-up over the classical model. Eventually, quantization has lead towards the foundation of various

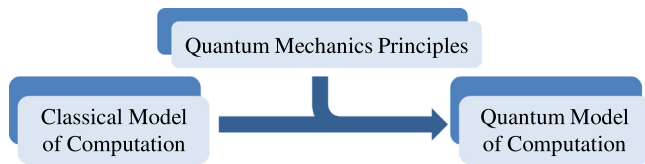


Fig. 1. Evolution of quantum walk.

quantum fields such as quantum computation, quantum information processing, quantum automata and quantum algorithms that are closely correlated.

In the past few decades, classical random walks and Markov chains have been used as a framework for developing algorithms in computer science and mathematics. Combination of theoretical physics principles with classical random walk provide a new paradigm, *i.e.* quantum walks, with surprising computational power and broad applications in the fields of mathematics, physics, biology, medicine, computer science and engineering.

Quantum walks, the quantum counterparts of classical random walks can be regarded as the most fruitful outcome/aspects of Quantum information theory as they form the basis of development for other models of computation, help in realizing the computational power of algorithms and also, motivates towards the development of a quantum computer that can solve the complex problems efficiently. Childs et al. [2] have proposed a continuous-time quantum walk algorithm for graph connectivity having exponential speed-up over any of the existing classical algorithms, while Shenvi et al. [3] developed a quantum algorithm having quadratically faster search times for finding a marked vertex on a hypercube. Many other quantum algorithms such as element distinctness, graph similarity search, quantum transport, oracle search, and others providing speed-ups over classical algorithm have been developed so far [4–6]. However, the authors have not given due importance to the trade-off between computational power and the increased complexity in developing these algorithms.

Another important aspect to consider is the transition between the quantum random walk and its classical counterpart [7, 8]. Furthermore, discrete-time quantum walk appears to offer added degrees of freedom due to the presence of a quantum coin over time-dependent Hamiltonian evolution in case of the continuous-time quantum walk.

In recent years, many researchers have published review articles relating to quantum mechanics, algorithms and quantum walk [9–11]. Most of the articles have been reviewed from a physics and mathematics perspective. In this article, we will provide a brief account of the evolution of quantum walks from classical random walks from the computer science perspective. It will enable the reader to have a better understanding of the quantum walk concepts without being overburdened by the underlying mathematics. We will present the necessary background of physics and computer science needed for better understanding of quantum walk concepts. Furthermore, we will provide a detailed view of the wide application domains of quantum walk with an insight into the recent developments in different fields.

1.1. Motivation

An account of major motivational factors for this article is as follows.

- (i) Introducing quantum physics principles in information processing has produced more efficient algorithms. Quantum algorithms have shown polynomial [12] and quadratic [13] speed-ups over classical algorithms. Quantum walk acts as

the core for designing quantum algorithms. So, adequate importance should be given to the exploration of infusing of quantum walks in algorithms.

- (ii) Quantum walks provide an intuitive framework for processing of quantum information. Ranging from development of new quantum algorithms to quantum information processing, quantum walks find utilization in every sphere of quantum computation. Different types of quantum walk find use in quantum processing based on the type of problem formulation. Therefore, realizing the usage and efficiency of quantum walks in varied domains is of utmost importance to today's technology evolution era.
- (iii) Recent review work in the field of quantum walks adheres to the mathematical modeling and simulation of various types of quantum walks. No detailed account of application domains and comparative study is provided in the literature which is very much required to broaden the horizon for application of quantum walks.
- (iv) This review may help researchers to apply quantum walk framework in quantum information processing and designing quantum algorithms.

1.2. Contribution

- (i) In this article, we will discover the underlying concepts of quantum walks from both physics and computer science perspective. This review of quantum information science may further allow readers to discover apropos ideas and concepts such as quantum information processing, quantum computation models, quantum circuits, quantum algorithms, speed-ups and decoherence.
- (ii) We also provide a comparative analysis of the existing review articles in the literature with our survey in Section 1.3.
- (iii) Taxonomy of various formulations of the quantum walk is presented in this review article along with their application areas and limitations. Further, a comparative study of classical random walk and quantum walk based on their application domain and complexity index is presented in Section 4.
- (iv) A comprehensive review of widely researched application areas and quantum walk models categorization is done in Section 5. It may help readers to get a clear understanding of the mapping of different quantum walk models to the appropriate application domain.
- (v) A brief account of the currently used simulators for visualization and implementation of quantum walks on graphs, and their comparison is provided in this article Section 4.7. Various coin operators and their mathematical models are discussed in Section 4.6.
- (vi) An insight into physical realization of quantum walk models and effect of decoherence on quantum walk dynamics is also provided in Sections 4.8 and 4.9, respectively.
- (vii) We also discuss open challenges and future research directions relating to quantum walk on graphs in Section 6.

1.3. Comparison with other surveys

There have been related attempts and literature surveys in the past on quantum walks, with their advantages and limitations. In this subsection, we overview such attempts and point-out the contrast between existing surveys and our survey. For example, Portugal [10] has discussed in detail the fundamentals of coined quantum walks on line and graphs, along with the staggered and Szegedy's model of quantum walk. A detailed survey on continuous time quantum walks on complex networks such as fractals,

deterministic and statistical networks have been presented by Mülken and Blumen [14], in which they also discussed relation between continuous time random walks and their quantum counterparts. An application oriented approach was covered and finely expressed with more application examples and algorithm designs using quantum random walks in surveys by Ambainis [15] and Santha [16]. Authors have provided a wide range of quantum random walk algorithms relating to spatial search problems, element distinctness and group commutativity among others and the parameters governing walk dynamics. The survey's by Ambainis [17] and Reitzner et al. [18] have provided detailed analysis on quantum walk evolution on line and graphs along with its application in varied domains as searching hypercube, grid, triangle finding etc. and constraint satisfaction problems respectively.

With respect to physical implementation, Montanaro [11] has briefly explained the various mediums and technologies such as bulk optics, NMR etc. These have been used effectively for physical realization of quantum walks. A detailed survey underlying the effects of noise and decoherence has been demonstrated by Kendon [19] along with the impact of coin dephasing and position decoherence on walk evolution on line. Konno [20] has also provided a detailed analysis of the disordered case of quantum walks on a line, cycle and trees. Perhaps the most closely related surveys to our work have been presented by Kempe [21] and Venegas-Andraca [22]. The surveys referred to the general analysis of quantum walks such as quantum walk types, application domains, evaluation parameters and disorder in quantum walk evolution. Our survey captures the comprehensive study and up to date review with additional insights into various quantum walk formulations referred to as completeness, simulation environment and discussions about the latest trends in application areas of quantum walks which were briefly discussed and/or missed in the existing studies. In Table 1, we summarize these aspects in comparison to the above-mentioned survey reports and contribution in literature.

1.4. Organization

The rest of the article is organized as follows. A detailed description about the background and evolution of quantum walks from classical random walks is given in Section 2 along with underlying mathematical concepts. It also includes definitions of the parameters considered for evaluating complexity of various quantum walk models and major differences between the classical and quantum walk. Section 3 explains various research questions and the review strategy followed in our article. Section 4 presents the categorization of quantum walks based on their evolution type, underlying graphs, number of quantum walkers and the type of quantum system. An account of the strengths and shortcomings of quantum walk models based on an application perspective is also explored in Section 4.5. Also, various coin operators, physical realization of quantum walk models, quantum simulators and effect of decoherence on quantum walk evolution are discussed in various subsection of Section 4. A description of the various application domains and sub-domains where quantum walk models are being utilized is discussed in Section 5. This section also contains a comparative study between quantum walk and its classical counterparts from a computational complexity perspective. Section 6 provides a discussion on various open challenges and future research directions. Finally, Section 7 concludes this paper.

2. Background

This section provides the prerequisites for understanding quantum walk models.

2.1. Classical random walk

Mathematically, a random walk is a process that describes a path consisting of a sequence of random steps in space. A random process is often described as a Markov process in which the next position of the walker depends only on its current position and does not depend on previous positions of the walker. The walker occupies discrete or finite positions and the randomness in the process is due to the transition of the walker between these states. In 1905, Pearson [24] introduced the concept of the random walk to calculate the probability of finding a drunken man after 'n' units of time and the probability was found to be highest at the origin.

2.1.1. Types of classical random walk

Classical random walks are classified in the following categories: discrete-time, continuous-time, recurrent, correlated, branching, and transient. Fig. 2 depicts the broad classification of classical random walk.

- (a) Discrete-time Random Walk (DTRW): If time is measured in discrete time steps, then the probability of finding the walker at a particular position 'i' after 't' time steps is given by the discrete-time random walk. A walk in discrete-time is defined by a chain of random variables $(X_t) = (X_1, X_2, \dots, X_n)$.
- (b) Continuous-time Random Walk (CTRW): If the time used to measure the probability at each position is a continuous function, i.e. in the form of a wave, then it is continuous-time random walk such as Brownian motion. In CTRW, the probability $P(X, t)$ that a process takes value X at 't' time steps is $P(X, t) = \sum_{n=0}^{\infty} P(n, t)P_n(X)$ [25]. Here $P_n(X)$ gives the probability that the process obtains the value X after 't' time steps, and $P(n, t)$ provides the probability of having taken n steps or jumps after time 't'.
- (c) Recurrent Random Walk (RRW): In RRW the walker is guaranteed to return to its starting position. This type of walk is feasible in one-dimension and two-dimensions only.
- (d) Transient Random Walk (TRW): In TRW, the walker has a positive probability of never returning to its starting position. It occurs in multidimensional space for dimensions greater than two.
- (e) Correlated Random Walk (CoRW): In CoRW, direction of motion of the walker at one instance of time is correlated with the direction of motion at the next time step. It is widely used in modeling the foraging behavior of animals.
- (f) Branching Random Walk (BRW): It is a stochastic process which generalizes the branching process and the random walk process.

2.2. Parameters to be considered in the evolution of classical random walk

Given an undirected graph $G(V, E)$, where 'V' and 'E' are the set of vertices and edges respectively, a random walk on graph G considers the following parameters to measure the efficiency of this walk.

2.2.1. Access time or hitting time H_{uv}

Consider a walker starting from node u . Hitting time H_{uv} is the expected count of steps to visit node v for the first time, starting from node u in a graph G [26].

Table 1
Comparison with existing surveys.

Reference	Completeness	Coin operators	Parameter coverage for complexity evaluation	Decoherence/Noise handling	Physical implementation	Simulation environment/Tools/Programming language	Application domains	Challenges/Open issues
Portugal (2018) [10]	*	*	**	†	—	—	*	—
Mülken and Blumen (2011) [14]	†	—	*	*	†	—	†	†
Reitzner et al. (2012) [18]	*	†	†	*	—	—	†	—
Ambainis (2008) [15]	*	†	†	—	—	—	*	—
Santha (2008) [16]	†	†	†	—	—	—	*	—
Montanaro (2016) [11]	*	—	*	—	✓	—	†	*
Ambainis (2003) [17]	*	†	†	—	—	—	*	*
Rong et al. (2013) [23]	†	*	†	**	✓	—	—	—
Venegas-Andraca (2012) [22]	*	*	**	*	—	—	*	—
Konno (2014) [20]	*	*	*	*	—	—	†	—
Kempe (2003) [21]	*	†	*	*	✓	—	*	*
Kendon (2007) [19]	*	*	*	**	✓	—	†	†
Proposed survey	**	**	**	*	✓	**	**	**

** : comprehensive analysis; * : general analysis; † : very little analysis; ✓ : considered; — : not considered.

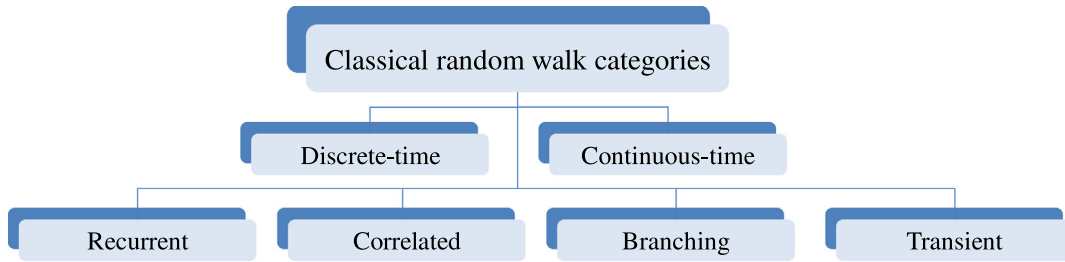


Fig. 2. Classification of classical random walk.

2.2.2. Commute time $\mathcal{K}_{uv}$

It is the expected count of steps a walker takes for a round-trip between the nodes u, v of graph G and is given by:

$$\mathcal{K}(u, v) = H(u, v) + H(v, u) \quad (1)$$

2.2.3. Cover time $C_{i,n}$

It is the expected count of steps to reach every node of graph $G(n)$ starting from a given initial or starting node i [27]. If no start node is specified, then cover time is calculated for the farthest start node.

2.2.4. Mixing time

It is the expected count of steps before the distribution will be approximately uniform or reaches its limiting distribution. A random walk on G can be defined by recurrent applications of a stochastic matrix S , where $S = \frac{1}{k_u}$, if (u, v) is an edge $\in E$ and k_u is the degree(u), where $u \in V$. The distribution of the classical random walk, $R_t = S^t R_0$ will converge to a stationary distribution π_r independent of R_0 for a connected and non-bipartite graph G , where R_0 is the initial distribution [28]. For a regular graph, π_r is uniform over the nodes $\in V$ of the graph. The mixing time is given by:

$$\text{Mixingtime} = \min\{T : \forall t \geq T, R_0 : \|R_t - \pi_r\| \leq \epsilon\} \quad (2)$$

Also, the mixing time is closely related to the gap between λ_1 and λ_2 of the stochastic matrix S , where λ_1 and λ_2 are the largest and second largest eigenvalues of S respectively [28].

2.2.5. Mixing rate

It is defined as the measure of how quickly a walk reaches its limiting distribution.

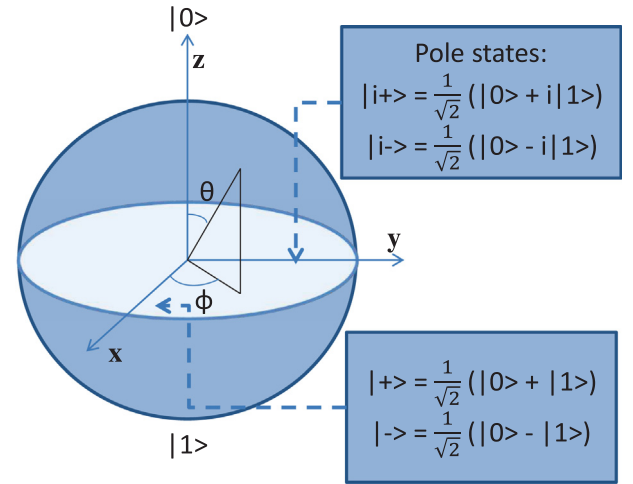


Fig. 3. Bloch sphere representation of qubit in two dimension.

2.3. Dynamics of the coined quantum walk model

Quantum walk is the quantum counterpart of a classical random walk. With the evolution of computation, quantum mechanics has been integrated into all aspects of computing for reducing the time complexity of the existing algorithms. It leads to a quadratic speed-up of some of the earlier classical algorithms. Quantum principles can be combined with other fields of science resulting in domains such as quantum information theory, quantum algorithm and quantum computation.

2.3.1. Preliminaries for quantum walk

This subsection provides insight into the mathematical formulation of the basic constructs which are essential for understanding the coined quantum walk and its time evolution.

- (1) **Qubit:** It is the term used for states in quantum computing [29]. A Bloch sphere representation of qubit in two-dimension is given in Fig. 3. A qubit $|\psi\rangle$ can either be in a basis state or it may be in a superposition of the basis states as defined by:

$$|\psi\rangle = x|\uparrow\rangle + y|\downarrow\rangle \quad (3)$$

$$p_{up} = |x|^2, p_{down} = |y|^2 \quad (4)$$

$$p_{up} + p_{down} = 1 \quad (5)$$

In the above equations $|\uparrow\rangle$ and $|\downarrow\rangle$ represent the basis states, and p_{up} and p_{down} gives the amplitudes of basis states.

- (2) **Quantum State:** In a given Hilbert space (H) spanned by a set of orthonormal basis states, $|k\rangle$ for $k \in \mathbb{N}$, the index $k \in \mathbb{N}$ is associated with a classical state after performing a measurement in the computation basis. A quantum state $|\alpha\rangle$ of the system is a unit vector in the H and can be described as

$$|\alpha\rangle = \sum_{k \in \mathbb{N}} a_k |k\rangle \quad (6)$$

In Eq. (6) $\sum_{k \in \mathbb{N}} |a_k|^2 = 1$.

- (3) **Superposition:** It is defined as the combination or addition of two or more basis states to yield another valid state. It may lead to constructive or destructive interference.
- (4) **Entanglement:** It refers to the relation between a pair or group of quantum particles such that the state of one particle cannot be reported independently of the quantum state of another, i.e. any action on one particle will subsequently affect the other, irrespective of their spatial distance. An example of an entangled two-electron spin state is $|\uparrow\rangle \otimes |\downarrow\rangle \pm |\downarrow\rangle \otimes |\uparrow\rangle$. Some basic aspects of entanglement include detection, characterization, distillation and quantification. This phenomena may manifest in compound quantum systems via entropic inequalities, Bell inequalities and quantum witnesses, etc. [30].
- (5) **Hilbert Space ($\mathcal{H}_t$):** It is the combination of coin Hilbert space and position Hilbert space. $\mathcal{H}_t$ for a discrete-time quantum walk can be described by:

$$\mathcal{H}_t = \mathcal{H}_p \otimes \mathcal{H}_c \quad (7)$$

In Eq. (7) $\mathcal{H}_p$ is the position space or the walker sites, $\mathcal{H}_c$ is the coin space, and $\otimes$ is the Kronecker or tensor product. However, in the case of continuous quantum walk the total Hilbert Space is the same as the $\mathcal{H}_p$ position space.

- (6) **Unitary operator ($\hat{U}$):** This type of function operates only on Hilbert space and preserves the inner product property and obeys:

$$U^\dagger U = U U^\dagger = I \quad (8)$$

In Eq. (8) $U^\dagger$ is the transpose complex conjugate of U , and I is the identity matrix.

The operator to be used for the time evolution of quantum walks must be unitary, i.e. the past and future position upon application of the operator must be unique. It is possible only if no data is lost in the transition between states of the system. The unitary operator outputs amplitudes instead of probabilities upon application to the

system wave function [31]. The system wave function after 't' repetitions of the unitary operator is given by:

$$|\psi\rangle_t = \hat{U}^t |\psi\rangle_0 \quad (9)$$

In Eq. (9) ψ_t is wave state vector after 't' steps, $\hat{U}^t$ is the unitary operator after 't' repetitions, and ψ_0 is the initial wave state vector defined as $|\psi\rangle_0 = |\text{position}\rangle_0 \otimes |\text{coin}\rangle_0$.

- (7) **Observables:** Many of the advantages that quantum walk has over its classical counterparts are a result of interference effects in quantum walks. These interference effects between walker and coin occur due to numerous applications of unitary operator $\hat{U}$ during the walk evolution. In order to know the dynamics of quantum walk, measurement is required at some point of the walk evolution. For that, a set of observables is defined in accordance to the basis states used to define the coin and quantum walker. Venegas-Andraca [22] suggested some methods to extract information from composite quantum systems by doing sequential measurements on coin and position sites of the quantum walker using the observables defined in Eqs. (10) and (11), respectively.

$$\hat{M}_{\text{coin}} = \beta_0 |0\rangle_c \langle 0| + \beta_1 |1\rangle_c \langle 1| \quad (10)$$

$$\hat{M}_{\text{pos}} = \sum_i \beta_i |i\rangle_p \langle i| \quad (11)$$

2.4. Parameters governing quantum walk dynamics on graphs

A quantum walk evolution whether discrete or continuous on a graph $G(V, E)$ has the following parameters.

2.4.1. Quantum hitting time

Time needed by a quantum particle to reach the target node as determined by a measurement of the target node is known as quantum hitting time [32]. The time required to detect or find a target node in a classical random walk or Markov chain is equivalent. However, hitting time for quantum walks is subtler to define, and there may be a difference in the time complexity of detecting and finding a target node in a graph [33].

2.4.2. Quantum commute time

Commute times of continuous-time quantum walk on a graph are analytically and experimentally explored by Emms et al. in [34] using Gauss–Laguerre quadrature. Quantum commute time on a graph is the time required by walker to travel to a node and back starting from some initial node.

2.4.3. Quantum mixing time

It is the minimum time after which the average probability distribution of the quantum system becomes approximately similar to its limiting distribution [35]. Although the limiting distribution does not exist for the quantum case, however, for a set of nodes, it can be defined as a long time average probability distribution of finding the quantum walker at each node.

2.4.4. One-shot hitting time

One-shot hitting time is applicable only to quantum walks with one final measurement, i.e. no intermediate measurements are allowed. The definition of one-shot hitting time was introduced by Kempe [36]. A quantum walk W has a (T, q) , one-shot $(|\psi_0\rangle, |u\rangle)$ hitting time if the probability to measure $|u\rangle$ at time T starting initially in $|\psi_0\rangle$ is larger than p , i.e. $\|\langle u|U^T|\psi_0\rangle\|^2 \geq p$ [36].

2.4.5. $|u\rangle$ - Stopped walk

The $|u\rangle$ stopped walk starting with initial state $|\psi_0\rangle$ is a procedure defined as an iteration of measurement with two projectors $\pi_0 = \pi_u = |u\rangle\langle u|$ and $\pi_1 = 1 - \pi_0$.

If π_1 is measured, then it states an application of quantum walk W , and

if π_0 is measured, then it means the process is stopped [37].

2.4.6. Concurrent hitting time

Concurrent hitting time is applicable to walks with partial measurements or absorbing boundaries at each step of the walk. The definition of concurrent hitting time is defined by Kempe [36] and Ying and Ying [38] as a walk W has a (T, q) , concurrent $(|\psi(0)\rangle, |u\rangle)$ hitting time if $|u\rangle$ - stopped walk from W and initial starting state $|\psi_0\rangle$ has a probability $\geq p$ of stopping at a time $t \leq T$.

2.5. Quantum walk operations

Two basic operations, namely unitary evolution and measurement can be applied to a quantum system. These operations are described in detail in this section, along with non-unitary evolution and the effect of combining two quantum systems.

2.5.1. Unitary evolution

In any quantum walk, the evolution of quantum states should strictly follow the unitary operator and its properties, as described in the previous section 8. A quantum state $|\alpha\rangle$ is mapped to the state $U|\alpha\rangle$, where $U^\dagger U = UU^\dagger = I$.

2.5.2. Measurement

A measurement in the computational basis of a quantum system whose quantum state is $|\alpha\rangle$ outputs an element $k \in \mathbb{N}$ with a probability $|\langle k|\alpha\rangle|^2$. Furthermore, the new state of the quantum system is $|k\rangle$ [28].

2.5.3. Non-unitary evolution

The unitary model is not the most usual model of quantum computation. Indeed, most commonly a quantum state is represented by ' ρ ', i.e. density matrix which is a semi-definite positive matrix having trace one. The density matrix of a quantum state $|\alpha'\rangle$ is $|\alpha'\rangle\langle\alpha'|$ and follows unitary evolution as per $U\rho U^\dagger$ and non-unitary evolution as per $\mathcal{E}(\rho)$, where $\mathcal{E}$ is the super-operator [28]. In other words, non-unitary evolution of a quantum system can be described as simply adding some qubits to the quantum system, then performing the required unitary transformation on the entire system and finally, discarding the extra qubits.

2.5.4. Combining two quantum systems

If two quantum systems A and B are represented using Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$ respectively, then their joint system can be described as Kronecker or tensor product of the Hilbert spaces as $\mathcal{H}_A \otimes \mathcal{H}_B$. If systems A and B are spanned by the basis states $|a\rangle$ and $|b\rangle$ respectively, then the joint system is spanned by basis states $|a\rangle \otimes |b\rangle$ and often abbreviated as $|a, b\rangle$ for the state $|a\rangle \otimes |b\rangle$.

2.6. Why the need of transition from classical random walk to quantum walk

Kendon and Sanders [39] have described the relationship between complementarity property and quantum walks. The study shows that quantum walk interposes between the classical random walk and the "wave walk" depending on the measurement strength of quantum walker's coin state. The spreading behavior of the quantum walk is way faster than the classical random walk.

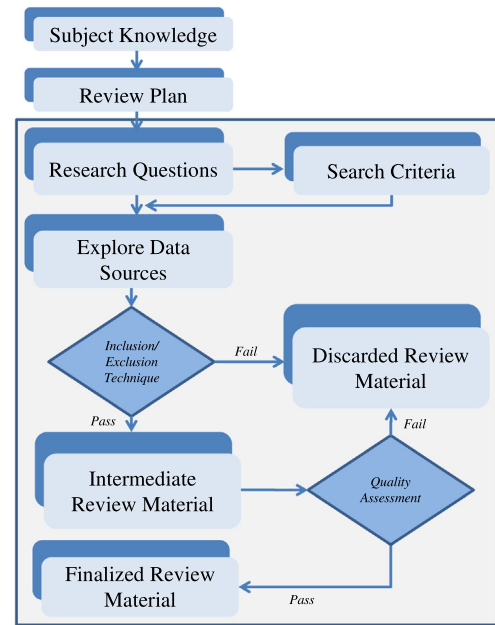


Fig. 4. Flowchart depiction of review method.

In quantum systems, a walker explores all of its feasible paths simultaneously leading to a speed-up in the spread of quantum walk. In classical walk, a walker can only choose to move towards one path out of all possible paths [33,40]. The matrix used in the evolution of quantum walk is unitary matrix as opposed to the stochastic matrix used in classical random walk. It implies that the evolution of quantum walk is reversible. Moreover, the position of walker is measured in terms of probability in classical walks as opposed to the position amplitude measurement in quantum walks.

Also, quantum walk possesses non-ergodic behavior while a classical random walk may be ergodic as in Markov-chain or non-ergodic as in the case of bipartite graphs. Thus, quantum walks do not converge to limiting distribution at any point during their evolution. Quantum walk differs from classical random walk in several properties such as, constructive and destructive interference which results in unique behavior of quantum walks. Furthermore, the randomness in classical walk arises due to transition from one state to any of the neighboring states, while in quantum walk randomness is solely attributed to the measurement operation on the quantum system.

3. Review method

The systematic review of quantum walk application on graphs is based on the proposed guidelines of Kitchenham et al. [41–43]. A systematic flowchart of the review method applied is demonstrated in Fig. 4

3.1. Review scheme

The inception of this review study begins with the formulation of research questions, elimination of trivial questions and exploration of databases to discover and evaluate the information. The review process has been divided into sub-phases that includes identification of primary studies, follow through quality assessment, extracting relevant data and finally, synthesizing the extracted data. The review process adheres to a pre-defined quality review protocol to minimize the reviewer's bias. In this protocol, the initial review is done individually, and the outcomes are finalized after joint discussions.

Table 2
Research questions and target objectives.

	Research questions	Target objectives
1	How does quantum walk evolution on graphs occur? Why is the transition from classical random walk to its quantum counterpart required?	It explores the quantum mechanics concepts and random walks, background behind the quantum walk, its types and implementation on graphs.
2	What are the different parameters to be considered for measuring effectiveness during the evolution of classical random walk and quantum walk on graphs?	It is anticipated to provide better understanding of the various parameters used to evaluate the evolution of walk on graphs and differences in the parameters considered for classical random walk and quantum walk.
3	Discuss the various quantum walk models or formulations applied in quantum walk implementation on graphs?	The question aims to explore the various quantum walk models/ formulations and their application in quantum walk evolution on graphs.
4	What are the various simulators available for visualization of quantum walk on graphs? Where can they be applied and what are their limitations?	It serves the purpose of exploring different simulators that aid in the visualization of quantum walk evolution on graphs. It helps to understand the simulation environment and the limitations of each simulator.
5	What is the impact of decoherence on quantum walk evolution?	It aims to presume the impact of decoherence and ways to handle noise in quantum walk evolution.
6	What are various formulations of quantum walk depending on its evolution using different unitary operators? What are the strengths and limitations of existing formulations?	It aims to present the different formulations of quantum walk based on the unitary operator it uses for its evolution on graphs. It will provide an understanding of the mapping between the unitary operator and quantum walk categories. It fulfills the purpose of describing strengths and limitations of different formulations.
7	What are the methods to obtain physical implementation of the quantum walk?	It captures the methods of quantum walk evolution in physical systems.
8	Discuss the various types of coin operators used in discrete-time quantum walk simulation? What are the strengths and limitations of these operators?	It provides a detailed account of the different coin operators used by discrete-time quantum walk in its evolution as per the given graph. It will describe the strengths and limitations of each of the coin operators to enhance the application of suitable coin as per the graph being used.
9	Discuss the various application domains of the quantum walk.	This question describes of various domains to which quantum walk can be applied along with key research findings in each domain. It will help in better understanding the application of quantum walk to various fields of science and engineering.
10	What are the various open challenges in the application of quantum walk on different types of graphs? Discuss the different research opportunities in quantum walk implementation on graphs.	It explores the existing research issues relating to the application of quantum walk in various domains and provides prospective researchers with future directions in the field of quantum information processing.

3.2. Research questions

This review focuses on the recognition and classification of existing literature on a quantum walk implementation on graphs, application of quantum walk in various domains and restrictions imposed its implementation and simulation of quantum walks. The research questions required for a systematic review on the quantum walk on graphs are listed in Table 2.

3.3. Data elicitation

A broad stance of the reviewer is a necessity to cover the substantial reporting of literature. Scholarly digital libraries provide an appropriate platform for finding relevant information from the vast pool of publications. Kitchenham et al. [43] and Brereton et al. [41] suggested the incorporation of a wide range of electronic data sources in information discovery. An extensive search of online digital libraries such as Physical Review Journals, Springer, ScienceDirect, ACM Digital Library, IEEE eXplore and Elsevier have been done for this purpose.

Other Sources: A comprehensive outlook is provided to the review by extending the scope of literature coverage. Apart from including the above mentioned digital sources, many other data sources such as books, technical reports, review letters and online literature apposite to this review are also included [42,43].

3.4. Search criteria

Almost all the searches performed has the keyword “quantum walk” and at least one of the keywords mentioned in Fig. 5 in the abstract. Look-up of relevant information to this review is broad in terms of coverage, period and quantity. Although a deliberate attempt was made while exploring search strings to ensure extensive coverage, some research articles have not

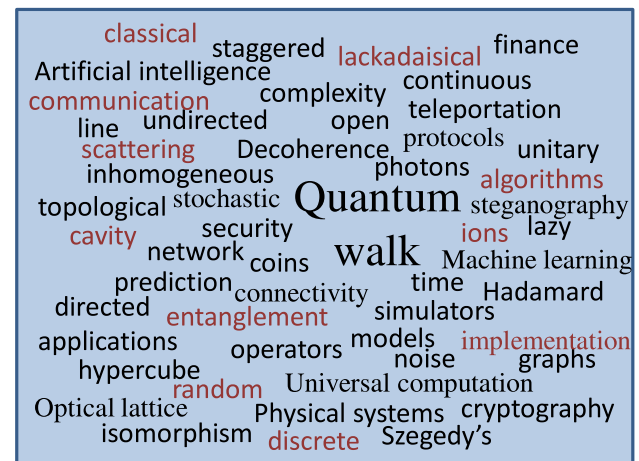


Fig. 5. Search strings used in search criteria.

been considered for various reasons, such as the search string considered in Fig. 5 is missing in the title or abstract of the article. Research community considers the terms “quantum walk” and “quantum random walk” equivalent, as the walk under consideration, is random in nature. An attempt to identify such papers has been made by manual searching for the references of papers with the given keywords in digital repositories.

3.5. Inclusion and exclusion approach

Quantum walk has wide applicability in various domains, as such the search string “quantum walk” yielded a large number of irrelevant papers. Article filtration proved to be difficult during

Table 3
Quality assessment questions.

	Question illustration	Answer
1	Does the research article refer to quantum counterparts of a classical random walk? The articles include scholarly coverage of quantum walk on graphs.	Yes
2	Are papers not related to quantum walks or random walks excluded from the literature survey?	Yes
3	Any of the research articles do not consider quantum walk evolution on graphs?	No
4	Do research article title, abstract, or full text provide a description about quantum walk principles and graphs?	Yes
5	Does the paper content cover quantum walk concepts?	Yes

the online digital library search. So initially, exclusion of unrelated articles is done based on the paper title, abstract and its full text. To provide a rich database, research articles from the onset date of a digital library to 2020 are included. Also, other review letters, research articles, books, tools, technical reports and other online sources are included for extensive literature coverage, completeness and comprehensiveness of this review. Inclusion and exclusion methodology was applied systematically to research articles in five stages, as presented in Fig. 6. Each stage comprises of three parameters, count of input research papers, filtration method applied and the count of output research papers after filtration.

3.6. Quality assessment

As the domain of this review article is diverse, many conferences, journals and magazines covering related articles were studied. Then, a systematic quality assessment has been performed on these articles as per the compliance standards of the Centre for Reviews and Dissemination (CDR) and Database of Abstracts of Reviews of Effects (DARE) [41]. The articles were selected as per the screening question listed in Table 3 and other high-quality research articles from reputed publications to extend the database of quality articles.

4. Quantum walk models

Quantum walk models represent the framework for the application of quantum walks in quantum algorithms and information processing [44]. Quantum walk models can be broadly classified into discrete-time and continuous-time and their sub-models as depicted in Fig. 7. A systematic growth timeline depicting various quantum walk models, and their applied fields is depicted in Fig. 8.

4.1. Discrete time quantum walk model (DTQWM)

DTQWM is the most widely used model of quantum computation. Discrete-time formulation was first discussed in 1993 by Aharonov et al. [45], in their seminal paper on quantum random walks. DTQWM evolves in discrete-time steps and is applied in discrete scope of positions. A wide set of notations exist for representing different quantum walk models [46]. In this review, we have selected one of the most common set of notations to keep the formulation easy to understand and consistent. For example, $\mathcal{H}_t$ denotes total Hilbert space, $\mathcal{H}_c$ represents a coin Hilbert space with its basis states, $|coin\rangle \in \mathcal{H}_c$ representing internal degrees of freedom and $\mathcal{H}_p$ represents position Hilbert space with its basis states, $|pos\rangle \in \mathcal{H}_p$ representing external degrees of freedom. $P(v, t)$ denotes the probability distribution over a set of states or

positions 'v' at time 't'. Similarly, $P(v, u, t)$ denotes the probability distribution of 'v' at time 't' with coin state 'u'.

We will provide a brief description of quantum walk on a line, followed by quantum walk evolution on a cycle. The section will finally conclude with a short review of quantum walk dynamics on higher dimensions, regular graphs and lattices.

4.1.1. Quantum walk on a line

Discrete-time quantum walk on a line is the simplest formulation of quantum walk model working on one-dimensional space, with two-qubit coin space, i.e. $|0\rangle$ and $|1\rangle \in \mathcal{H}_c$ and a set of 'v' position states $|pos\rangle \in \mathcal{H}_p$. The total Hilbert space $\mathcal{H}_t$ is denoted as Kronecker product of coin and position Hilbert spaces, i.e. $\mathcal{H}_t = \mathcal{H}_p \otimes \mathcal{H}_c$ and the basis of $\mathcal{H}_t$ is represented by vectors $|u, v\rangle \in \mathcal{H}_t$. $\hat{U}$ represents the unitary evolution operator of quantum walk. $\hat{U}$ comprises of coin operator $\hat{C}$ and conditional shift operator $\hat{S}$ on $\mathcal{H}_t$. The application of shift operation shifts the walker to the left or right depending on the output of the coin operator. One step of unitary operation can be represented as $\hat{U} = \hat{S}(\hat{I} \otimes \hat{C})$, where $\hat{I}$ is the identity matrix. A commonly used quantum coin, SU(2) operator [47] can be defined in its matrix form as

$$\hat{S}U(p, \alpha, \beta) = \begin{pmatrix} \sqrt{p} & \sqrt{1-p}e^{i\alpha} \\ \sqrt{1-p}e^{i\beta} & -\sqrt{p}e^{i(\alpha+\beta)} \end{pmatrix} \quad (12)$$

Here, p and $1-p$ denote the probability of quantum walker moving left or right respectively. The parameters α and β denotes the most general $\hat{U}$ operator. These parameters, i.e. (p, α, β) are selected at the starting of the quantum walk, and they remain unchanged during the walk evolution. A suitable choice for coin operator is Hadamard coin operator which is obtained by substituting the values of $p = 1/2$, $\alpha = \beta = 0$.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (13)$$

The Hadamard coin operations are defined using superposition principle and given by the following equations:

$$|+\rangle \text{ or } \hat{H}|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (14)$$

$$|-\rangle \text{ or } \hat{H}|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (15)$$

Fig. 9 represents DTQWM on a line based on the outcome of the coin operator.

The shift operation is performed on the quantum system following the equations:

$$S|v, 0\rangle = |v+1, 0\rangle \quad (16)$$

$$S|v, 1\rangle = |v-1, 1\rangle \quad (17)$$

$$\hat{S} = \left(\sum_v |v+1\rangle\langle v|_p \otimes |0\rangle\langle 0|_c \right) + \left(\sum_v |v-1\rangle\langle v|_p \otimes |1\rangle\langle 1|_c \right) \quad (18)$$

Eq. (18) represents one step of the quantum walk. The quantum state after t steps of the walk corresponds to application of unitary operation t times:

$$|\psi\rangle_t = (\hat{U})^t |\psi\rangle_0 \quad (19)$$

Mathematical derivation of quantum state for steps 0, 1 and 2 are given by [37]:

$$|\psi\rangle_0 = |0\rangle_p \otimes \frac{1}{\sqrt{2}}(|0\rangle_c + |1\rangle_c) \quad (20)$$

$$|\psi\rangle_1 = \frac{1}{\sqrt{2}}|1\rangle_p|0\rangle_c + \frac{1}{\sqrt{2}}|-1\rangle_p|1\rangle_c \quad (21)$$

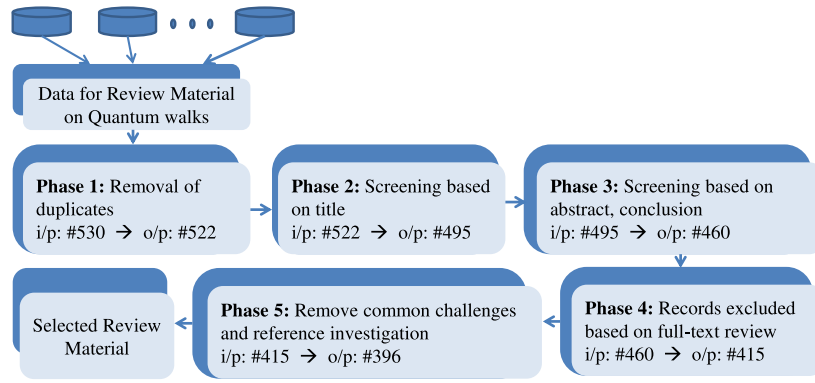


Fig. 6. Flowchart for search criteria depicting selection and inclusion procedures.

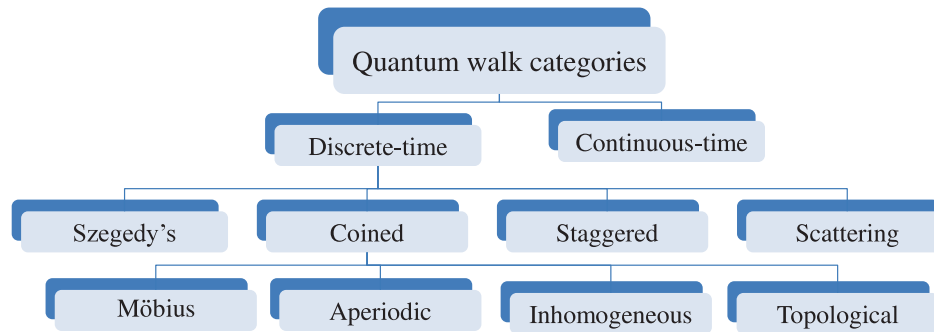


Fig. 7. Broad Categorization of quantum walk models.

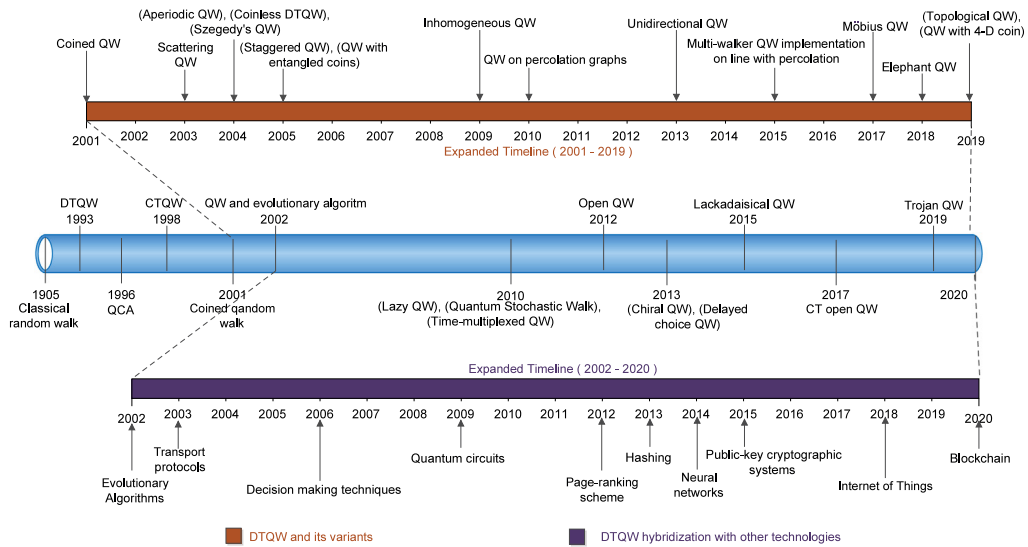


Fig. 8. Timeline of various quantum walk models and their applied fields.

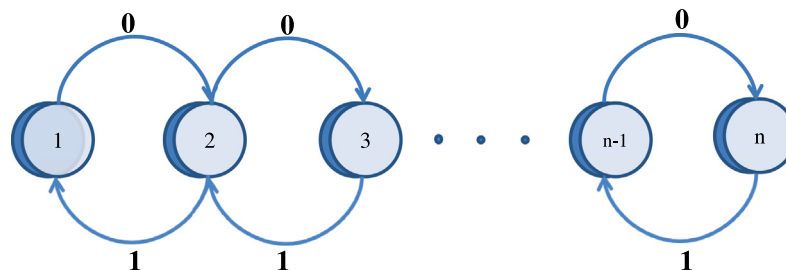


Fig. 9. Discrete-time quantum walk evolution on a line.

Table 4
Probability of being at position 'j' after 'N' steps of quantum walk.

N \ j	-5	-4	-3	-2	-1	0	1	2	3	4	5
0						1					
1					1/2		1/2				
2				1/4	2/4		1/4				
3			1/8	5/8		1/8		1/8			
4		1/16		5/8	1/8		1/8		1/16		
5	1/32		17/32		1/8		1/8		5/32		1/32

$$|\psi\rangle_2 = |2\rangle_p \left(\frac{1}{2} |0\rangle_c + 0 |1\rangle_c \right) + |0\rangle_p \left(\frac{1}{2} |0\rangle_c + \frac{1}{2} |1\rangle_c \right) + |-2\rangle_p (0 |0\rangle_c - \frac{1}{2} |1\rangle_c) \quad (22)$$

Table 4 represents the probability distribution of quantum walk on a line.

DTQWM on the line has been rigorously studied in the past years by Aharonov et al. [28], Konno et al. [48], Rigovacca and Franco [49], Machida and Konno [50]. Two widely used approaches to study discrete-time quantum walk on line are Schrödinger and combinatorial approach.

Nayak and Vishwanath have analyzed unbiased quantum walk evolution on a line, specifically Hadamard walk and they proved that the probability distribution is uniformly spread over the interval of $[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]$ which leads to linear mixing of quantum walk [51]. Later in 2001, Ambainis et al. have extended the work in [51] by analyzing dynamics of Hadamard walk with absorbing boundaries and calculated the exit probability of a quantum walker to be $2/\pi$ and $1/\sqrt{2}$ in the presence of an absorbing wall immediately to the left of starting point of walker and an additional wall at a random position respectively [52]. Abal et al. [53] have explored the impact of initial conditions (local or non-local) on the degree of asymptotic entanglement. They acquired the entanglement value of 0.872 for local conditions and unrestricted coin states while its value varied from 1 (complete) and 0 (no entanglement) for non-local initial conditions. Romanelli et al. provided an analytical expression for quadratic rise in variance of walker's position with respect to time by dividing the master evolution equation into interference and Markovian terms [54]. Further, Romanelli et al. have studied the effects of decoherence on quantum walk evolution on line that may limit its implementation on physical systems. Decoherence is introduced in the system using simultaneous measurements of particle position and walk chirality, and due to broken or missing links. The results show an inverse relation between probability of breaking links and time at which evolution becomes decoherent [55]. Brun et al. provided analytical expressions for time-dependent moments of position in the presence of decoherent quantum coin in [8] with variance growing linearly in time, which is unlikely of unitary walk. In 2005, Inui et al. [56] have proposed a variant of Hadamard walk in which the evolution is governed by three internal degrees of freedom (move either left or right or stay at the same place) rather than earlier proposed two degrees (either left or right). As a result the quantum walker is trapped with a high probability in the vicinity of origin. Chandrashekar [57] proposed a technique to implement the Hadamard walk using Bose-Einstein condensate in which the Hadamard operation (rotation), is operated using rf pulse and unitary shift operation using the Raman transition scheme. Dür et al. [58] have experimentally realized one-dimensional quantum walk using neutral atoms which are trapped in optical lattices along with the impact of errors and decoherence induced in the system during evolution. Later in 2011, Xue et al. [59] realized the experimental implementation of a multi-step Hadamard walk using a single ion-trap and randomized Hadamard coin-flip phase.

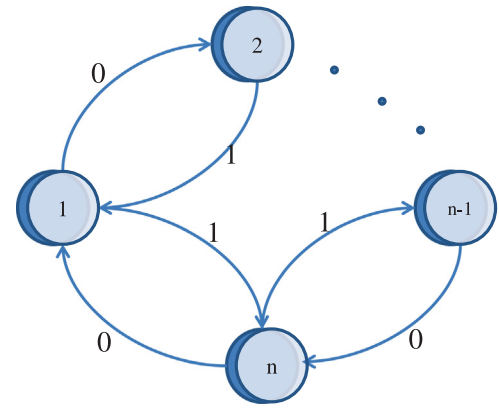


Fig. 10. Discrete-time quantum walk evolution on a cycle.

Konno [60] analyzed the dynamics of quantum random walk using $\hat{U}$, a 2×2 unitary matrix which is also used to obtain a combinatorial expression for n th moment of the walk. Furthermore, he extended his work in [60] and has provided mathematical expressions for calculating the sojourn time of Hadamard walk using generating function and path counting approach [61].

Jayakody and Nanayakkara [62] have introduced a decoherence model for a quantum walk on a line that uses concurrent application of bit-flip and phase-flip operators randomly on the coin state. Wong in [63] provided a numerical proof of ballistic spread, i.e. $O(n)$ for 'n' time-steps of quantum walk which is quadratic-ally faster than the diffusive spread realized for classical random walk. Tude and Oliveira [64] discussed the formal definitions of thermal equilibrium and temperature for a discrete-time walk with two and three-sided quantum coins. Ghizoni and Amorim [65] have investigated the entanglement and transport properties between position and spin of quantum walk on a line following Gaussian, i.e. delocalized state and delta-like, i.e. local state distributions.

4.1.2. Quantum walk on a cycle

Bednarska et al. [66] have mathematically explored the asymptotic dynamics of Hadamard walk on a cycle. An analytical formula for limiting distribution of Hadamard walk is provided for the case when the cycle length tends to infinity. Dukes in [67] has calculated the period P_k of a cycle C_k having k vertices for generalized quantum walks using the eigenvalue approach. The results provide values of period T_2, T_4, T_8 to be 2, 8 and 24 respectively for the Hadamard walk. Recently in 2017, Konno et al. have elaborated the work in [67] by proving analytically that Hadamard walk period holds only for T_2, T_4, T_8 case using the cyclotomic polynomials and path counting approaches [68]. Further in 2019, Kajiwarra et al. [69] extended the previous work using the two-state walk in [67] and [68] by analyzing the periodicity of the three-state walk on the cycle C_k using a novel cyclotomic field. The method is validated for Fourier and Grover walks, which showed that these walks only possess a period T_3 equal to 12 and 6, respectively.

Fig. 10 represents DTQWM on a cycle based on the outcome of the coin operator.

Arai et al. [70] have provided a spectral analysis for periodicity behavior of space in-homogeneous quantum walks in terms of isospectral and non-isospectral quantum coins. They extended the results for spectral coins while showing that the periodic behavior of general coin at the origin determines the periodicity of Hadamard walk in non-isospectral coin case. Álvarez et al. [71] presented a non-linear system of $|G| - 1$ equations

whose solution describes the complete set of co-cyclic Hadamard matrices over group G , where order of G is $4t$. Furthermore in 2020, Álvarez et al. [72] expanded their earlier work to describe co-cyclic Hadamard matrices (in [71]) evolved over a loop (specifically Goethals-Seidel loops). They also introduced the concepts of pseudo co-cycle, pseudo co-boundary and co-boundary over a quasi group and co-cyclic Hadamard matrices.

Georgopoulos and Zuliani [73] presented the implementation of DTQW with Hadamard coin on cycles with eight, sixteen, thirty-two and sixty-four nodes using controlled inversion approach and rotation operators. Although the inversion approach is simple, it leads to caveat in the form of ancilla registers. Rotational approach, on the other hand, overcomes this limitation but results in larger quantum circuit and slower computation. He et al. [74] have explored the coherence dynamics of quantum walk evolution on a line using two different scenarios in relation to an initial state. In the first case, the quantum walker starts from a single or pure position state while in the other case the initial state is an equal-superposition of all states. The coherence of the system quantifies as per the formula $(K - 1)$ and $(2K - 1)$ for any odd and even time-steps respectively, where K is the total number of sites in the system. In 2017, Moradi and Annabestani [75] introduced a novel quantum walk, namely Möbius Quantum Walk (MQW). MQW, in addition to coin and position Hilbert spaces, has an extra Hilbert space that enables the walker to rotate around its axis of movement. MQW is able to reach uniform distribution for any count of nodes in the system by fine tuning β (Möbius parameter) which cannot be affirmed for other quantum walks.

4.1.3. Quantum walk on higher dimensional graphs

There have been many implementations of DTQWM on a line using one or two quantum walkers. However, implementing DTQWM on general graphs using more than two walkers has always been a challenging task for researchers. But in 2010, Rohde et al. [76] presented a novel approach for analyzing and physically implementing the model on arbitrary graphs and with an arbitrary count of quantum walkers. The approach focuses on some essential properties of DTQWM such as distinguishability, difference between interference patterns of classical and quantum random walk, characterization and measurement. Štefaňák and Skoupý [77] have shown the perfect state transfer using two marked vertices, i.e. sender and receiver on complete bipartite graphs. They validated the transfer for two possible scenarios. Firstly, when both the marked nodes are in the same set of the partitioned graph. It resulted in perfect transfer as in star graphs and transfer dynamics remain independent of the other partition. Secondly, when the marked nodes are in two different partitions. In this case, fidelity value equal to unity is achieved if partitions have a similar size. Bai et al. [78] have proposed various quantum kernels using DTQWM on arbitrary graphs. A quantum kernel is defined as the negative exponential of Jensen-Shannon divergence, which is computed for density matrices obtained during discrete-time walk evolution on graphs.

Fig. 11 represents DTQWM on a higher dimensional graphs based on the outcome of the coin operators (DFT and Grover) discussed in Section 4.6.2, where $\alpha = 2\pi/3$.

D'Ariano et al. [79] investigated the quantum dynamics of Weyl's discrete quantum walk evolution on 3+1 dimensions. Further, an analytical solution to the walk evolution is also provided in position description by D'Ariano et al. approach. In 2019, Dai et al. [80] provided the spectral analysis of quantum walk using numerical simulation of DTQWM on Cayley graph of dihedral or non-Abelian group. Further, a comparison between quantum walk models with and without memory is also presented using quantum walk evolution on cycle and Cayley graph respectively. Bai et al. have proposed a similarity measure for weighted graphs

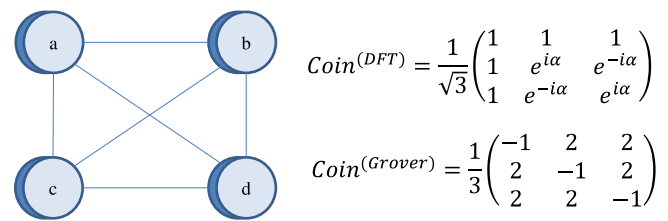


Fig. 11. Discrete-time quantum walk evolution on graphs.

in [81] using DTQWM, but since directly simulating discrete-time walk on a weighted graph is not an economically feasible solution. Therefore, a minimum spanning tree is first pruned from the weighted graph and then quantum dynamics of this MST are analyzed using DTQWM for similarity index using Perron-Frobenius operator. Chen et al. have proposed novel DTQW technique for efficient quantum communication adaptable to various general networks such as rail network, inverted-crown and butterfly network [82]. The proposed technique allows for perfect state transfer with probability and fidelity equal to unity between the initial and final qubits in finite steps, thus making it easier to execute. In 2020, Abd el-Latif et al. proposed a novel technique for cryptography by constructing efficient S-boxes (substitution boxes) using cascaded DTQWs and chaos stimulation in [83]. Also, an efficient Pseudo random number generator (PRNG) technique is presented to overcome the existing periodicity problem. Furthermore, an atypical image encryption technique is provided by combining the aforementioned techniques.

Based on the type of coin used and scattering behavior of quantum walk, Kendon [19] has proposed three different implementations for a quantum walk in discrete spaces such as graphs, i.e. generalization of diffusion like dynamics with continuous-time quantum walk, coined quantum walk and scattering quantum walk.

Discrete-time quantum walk models can further be classified into staggered, Szegedy's and scattering models, which are further described in succeeding subsections.

4.1.4. Staggered quantum walk model (StQWM)

Portugal et al. first described this model formalism in 2016 by focusing on proving the equivalence of Szegedy's and staggered model in spatial search problem. This model includes the majority of the subclass of coined or DTQWM, including Szegedy's model, where Szegedy's model can be viewed as an instance of staggered model [5]. Portugal [84] extended this work by simulating staggered walk model on graphs and proved that under the staggered formalism, coined and Szegedy's model are described for a special subclass of graphs which holds even for searching framework. Furthermore, Portugal and Fernandes [85] applied this model with Hamiltonians for efficient quantum search in two-dimensional lattice by using the graph tessellations and associated reflection operators as Hamiltonians. Coutinho and Portugal [86] provided a close association between the continuous-time and discrete-time quantum walk models by using staggered model evolution on Cayley graphs. This connection provides a way of discretization of continuous walk model and aids in uniform mixing and perfect state transfer in staggered model. Chagas et al. [87] have explored dynamics of staggered walk evolution on a hexagonal lattice, where unitary operator is the product of local operators (defined for each tessellation) and the walk evolution is defined over a group of overlapping tessellation that covers the edges of the underlying graph.

4.1.5. Szegedy's quantum walk model (SzQWM)

Szegedy [6] proposed SzQWM in 2004 as a quantum counterpart of classical Markov chain following the constraints of ergodicity and symmetry in the transition matrix. The proposed model was successful in achieving quadratic speed-up over its classical counterparts in searching for multiple marked elements. Magniez et al. [88] proposed a naive search algorithm using this model [6] based on amplitude amplification technique. Their approach involved using a phase estimation scheme on a quantum walk to simulate reflection operator, which in turn aided in amplitude amplification and also incurred lesser cost and larger search space. Tani [89] proposed an optimal algorithm based on SzQWM for finding a claw of constant size 'x' where $x > 1$ and the functions may have different sized domains. Loke and Wang [90] proposed an efficient theoretical quantum circuit implementation of Szegedy's model for various classes of graphs, including page-ranking algorithm also for special subclass of graphs. Furthermore, Higuchi et al. [91] investigated the periodicity of Szegedy walk on a certain class of graphs such as strongly regular graphs, complete bipartite graphs and complete graphs.

4.1.6. Scattering quantum walk model (ScQWM)

Feldman and Hillery [92] proposed a variant on DTQW evolution on finite general graphs from scattering theory perspective. Two nodes of the graph are selected, and one half line is attached to each of the nodes. ScQWM is then formulated to start from one half line and traverse through the underlying graph to reach the other half in 'n' time steps, whose probability is described in the form of transmission amplitudes for that graph. Gábris et al. [93] analyzed the effect of realistic noise and losses on ScQWM in a linear and passive optical network. While a uniform loss in all directions leads to search time re-scaling, its deviation leads to enhancement in search time efficiency. Tarasinski et al. [94] provided an effortless technique to calculate topological invariants (depending on system symmetry) in ScQWM on finite optical experiment even in the presence of disorders in the system. In recent work [95], Xue et al. achieved complete quantum speed-up in search efficiency by applying ScQWM on a Johnson graph $J(N, K)$ with $K = 3$. The graph $J(N, K)$ comprises of N symbols, where vertices are described by K -element sub-sets of these symbols with the constraint that two vertices are adjacent if and only if they differ in exactly one symbol. Furthermore, Komatsu et al. [96] provided a numerical expression for scattering operator while applying ScQWM on a one-dimensional lattice with impurities which in turn aids in transmissions and reflections. The mathematical expression is greatly dependent on the approach of counting paths of walkers in the quantum system.

4.2. Continuous time quantum walk model (CTQWM)

The origin of continuous-time quantum walk dates back to 1964 when Feynman et al. first described its evolution on a discrete lattice. However, their utilization in designing quantum algorithms came much later in 1998, where Farhi and Gutmann [97] used it for decision tree traversal. Fig. 12 represents CTQWM on graphs based on the corresponding adjacency matrix.

The CTQWM is defined by the unitary operator $\hat{U}$, given by Eq. (23) in which the Hamiltonian is derived from the graphs adjacency matrix.

$$\hat{U} = e^{-iHt} \quad (23)$$

In Eq. (23) H is the Hamiltonian operator. Furthermore, the quantum walk evolution may also include superposition of states which is related to the property of Schrödinger equation given by:

$$i\hbar \frac{\partial |\Psi(t)\rangle}{\partial t} = \hat{H} |\Psi(t)\rangle \quad (24)$$

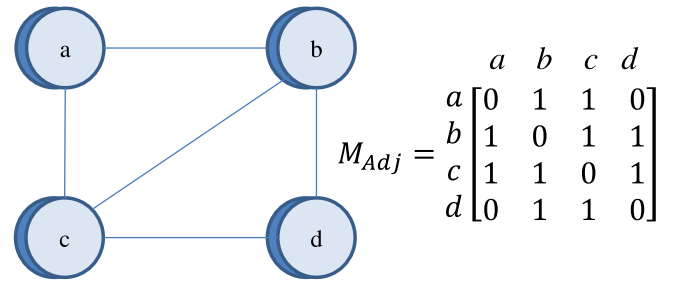


Fig. 12. Continuous-time quantum walk evolution on graph.

In this Schrödinger equation i is the imaginary number unit, $\hbar$ is the reduced Planck's constant defined as $\hbar = \frac{h}{2\pi}$, $\Psi(t)$ is state vector at time t in quantum system, and $\hat{H}$ is Hamiltonian operator. The current position of the walker $|\psi(t)\rangle$ after time 't', provided its initial position $|\psi(0)\rangle$ is given by:

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle \quad (25)$$

In continuous quantum walk, the position space is the same as total Hilbert space. A quantum walker lives in an infinite space but with countable dimensions $\mathcal{H}_p$ and is denoted by $|pos\rangle \in \mathcal{H}_p$ with canonical basis states $|i\rangle_p$ spanning $\mathcal{H}_p$ along with any superposition states of the form $\sum_i \beta_i |i\rangle_p$ subject to condition $\sum_i |\beta_i|^2 = 1$. In general, the initial state of the walker is defined at the 'origin', i.e. $|pos\rangle_{\text{initial}} = |0\rangle_p$.

4.2.1. CTQWM on the line

Konno [98] has provided a mathematical expression for weak limit theorem of continuous-time walk equivalents of discrete-time walk on line whose evolution is defined using Hadamard transformation. Further, he presented and validated the mathematical formulation for exact probability distribution of CTQWM on a line [99]. A central limit theorem for CTQWM on homogeneous tree has been proposed by Konno [100] using quantum probability theory. Recently, in 2020 Delvecchio et al. [101] presented an analytical investigation of the correspondence between CTQW evolution on a line and quantum kicked rotor at resonance (quantum) conditions. Further, a verification of the equivalence of probability distribution of both cases was provided for an appropriate kick-strength of quantum rotor.

4.2.2. CTQWM on general graphs

Jafarizadeh and Salimi [102] investigated the dynamics of continuous-time quantum walk evolution on various finite and infinite graphs such as cycle, line, complete and general graph using spectral distribution approach obtained through the associated adjacency matrix. Spectral distribution approach further aids in exploring the asymptotic behavior of amplitudes using the stationary phase approximation technique. Mülken et al. proposed the application of CTQWM on small world networks for quantum transport of excitons. Small world networks are formed when additional edges were introduced randomly in an existing one-dimensional ring structure. The findings reveal very fast quantum transport from one node to another in the network and no equipartition of the network is done during transition [103]. Agliari et al. [104] investigated the quantum transport phenomena in definite discrete-structures using CTQWM and calculating average displacement of quantum walker as a time-function. Using numerical and analytical methods, they proved that quantum walk performance is weakened by in-homogeneity and finiteness of the underlying substrate.

Varbanov et al. [105] presented various scenarios for infinite hitting time of continuous-time quantum walk on different

graphs and their dependence on symmetry of the underlying graph. Further, a mathematical expression is provided for relation between the quantum hitting time and measurement rate γ . Philipp and Portugal [106] provided the accurate simulation of coined quantum walk evolution on a generic graph using CTQWM on a larger graph (by expanding the original graph and alternatively switching edges on and off). Qiang et al. [107] explored the link between computational complexity and CTQW dynamics on arbitrary circulant graphs. Quantum circuits have been employed to efficiently sample the probability distribution of CTQW evolution on this class of graphs to describe supremacy of quantum walk over its classical counterparts. Wong [108] addressed the issue of spatial search result sensitivity to γ , i.e. jumping rate of the quantum walk. A spatial search is performed on the simplex of complete graphs, wherein the position of marked nodes alters the performance of quantum walk. Yalouz and Pouthier [109] proposed a tight-binding CTQWM using an exciton on an extended star graph, where the central node of this graph is occupied by a trap. Quantum dynamics of the exciton are governed by the absorption rate and the size of underlying star graph. An optimal quantum transfer occurred when the above said parameters are carefully chosen and it also leads to minimization of exciton absorption time. Benedetti et al. [110] investigated the quantum dynamics of CTQW evolution on dynamically percolated graphs in the presence of space and time-dependent noise. The results were validated on CTQWM on line and for spatial search performance on star and complete graphs.

Tang et al. [111] demonstrated the physical implementation of a two-dimensional CTQW to investigate quantum transport properties with the help of only external geometry of photonic wave-guide arrays. A 49×49 nodes photonic chip is derived from a large three-dimensional structure, which is constructed by using a femto-second laser writing technique. Chawla et al. [112] explored and compared the CTQW dynamics of a quantum walker on a square lattice and quasi-crystal lattices, both in the absence and presence of disorder due to disconnection between nodes. A major fraction of quantum state is found to be localized around the origin in quasi-crystal, but not in square lattice.

Osada et al. have investigated and analyzed quantum spatial search using CTQWM on the scale free Bollobás network. The analysis proves that the time required to concentrate the walker on marked vertex depends mainly on the global structure of the nodes surrounding marked vertex and very little on the local properties of the marked vertex such as its degree. However, due to its dependence on marked vertex properties, optimal quantum search requires some prior-knowledge about graph topology and marked vertex attributes [113]. Wang et al. have proposed a novel isomorphism mapping technique, namely IsoMarking for regular and symmetric graphs. The proposed scheme is based on CTQWs which are highly sensitive to underlying topological structure of the graph and even diminishes the harmful effects of symmetry [114].

4.3. Relationship between DTQW and CTQW

Classical and quantum random walks can both be described in terms of continuous and discrete-time walk models. Although in classical case continuous-time model can be obtained from discrete model as a limit of series of the latter model. But in quantum case, these two models vary fundamentally owing to extra degrees of freedom in discrete model. Several attempts have been made in the past recent years to find the exact relation between the two models. Strauch [115] presented the precise mapping relation between the two models classically (diffusion) and quantumly (wave propagation) by simulating quantum walks both in one-dimension and also in higher dimensions. Childs [116]

presented an accurate correspondence between continuous and discrete-time quantum walk models on general graphs. This correspondence lead to efficient simulations of even non-sparse Hamiltonian in which was difficult to achieve earlier. Furthermore, Childs [117] proved continuous-time quantum walks to be universal computation primitive by simulating the sparse Hamiltonian corresponding to adjacency matrix of some low-degree undirected graph. It included the implementation of a set of quantum gates using scattering method. Later, Lovett et al. [118] implemented the same set of quantum gates using discrete-time model, thus proving both quantum walk models to be universal computational primitives. D'Alessandro in [119] proved the existence of continuous-time walk dynamics, i.e. Hamiltonian as a limit case of its discrete-time counterpart on the same lattice or regular graph. Schmitz and Schwalm [120] proposed a DTQW that approximates the dynamics of a CTQW evolution on general graphs by using a coin operator that exhibits similar dynamics as that of continuous-time walk evolution. Recently in 2020, Funakawa et al. [121] have constructed the time-operators using different examples for both discrete and continuous-time quantum walks and have also provided a way to measure their spectra and deficiency indices.

4.4. Miscellaneous models

In this section, we provide a small introduction to the various quantum walk models. Also, we briefly discuss the recent developments and work done in these models.

4.4.1. Quantum stochastic walk model (QSWM)

Whitfield et al. introduced the concept of QSW in 2010 [7] where they demonstrated QSW as a generalized case of classical random walk and quantum walks using the axiomatic approach. The QSWM is governed by the Kossakowski Lindblad master equation defined as follows:

$$\frac{d}{dt}\rho = \mathcal{L}(\rho) = -i[\hat{H}, \rho] + \sum_k \gamma_k (\mathcal{L}_k \rho \mathcal{L}_k^\dagger - \frac{1}{2}(\mathcal{L}_k \mathcal{L}_k^\dagger), \rho) \quad (26)$$

In Eq. (26) ρ is the density operator of a quantum walker, $\hat{H}$ is the Hamiltonian derived from adjacency matrix of the underlying graph and $(\mathcal{L}_k, \mathcal{L}_k^\dagger)$ are the Kossakowski Lindblad operators with γ_k as their corresponding transition rates. A complete set of protocols is defined for all three walk models governing the transition of a walker from a vertex to its neighbors. A generic diagram representing the association between QSWM, DTQWM and CRW model is represented using Fig. 13, where ω is the dissipative coefficient.

The rules are validated using walk evolution on a line and a three-level glued tree. Many authors have provided numerical validation of QSWM, but its simulation has many limitations in terms of computational resources. Govia et al. [122] used quantum trajectories style approach to simulate QSWs and open quantum systems on a coherent quantum computer with minimum resources. Falloon et al. in 2017 [123] developed a Mathematica package, namely QSWalk to simulate QSWs on arbitrary graphs. Furthermore, in 2019 Glos et al. [124] developed a similar functionality software package in the Julia programming language for analyzing the simulation of QSW and open quantum system on directed arbitrary graphs.

Human decision-making experiments have reported violation of fundamental laws of total probability during human information processing and reaction to input data. As such, quantum information theory has been found to be more closely related to the dynamics of real-world decision making than its classical counterpart. Martínez-Martínez and Sánchez-Burillo [125] provided a quantum cognitive model based on QSW evolution on

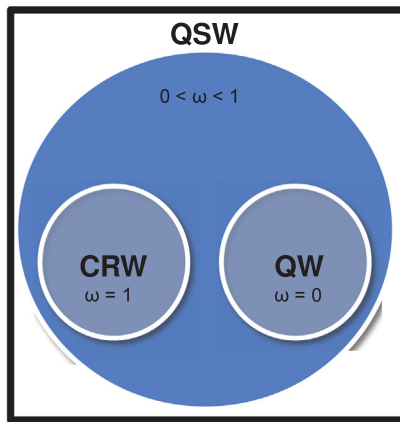


Fig. 13. Quantum stochastic walk model generic representation.

networks that incorporates quantum theory laws for a more realistic human-like decision making. Domino et al. [126] analyzed the properties of QSW evolution using the parameter asymptotic scaling exponent or mean-squared distance traveled by the walk as a way to measure the degree of walk coherence. Their proposed scheme also allows for global dissipation behavior of the walk, while keeping its ballistic propagation intact. Furthermore, Domino et al. extended their work in [126] to analyze the QSW dynamics on a directed arbitrary graph using super-diffusive evolution on a line. But, due to its environmental interaction, the evolution suffers from spontaneous moralization. To overcome this effect, a correctional scheme is provided to preserve underlying graph topology and prevent unnecessary transitions in graph traversal while preserving the super-diffusive characteristic simultaneously [127]. Tang et al. [128] efficiently simulated two major components of Hopfield neural networks, namely associative memory and its firing pattern using quantum stochastic walk evolution on a three-dimensional photonic chip. Pozza and Caruso addressed the problem of quantum state discrimination in their recent work [129], where they provide different QSWMs for the same. QSW dynamics for an open system evolution on a network is analyzed and network topology weights are numerically optimized to give a solution to the aforementioned issue which ultimately results in a higher probability value for the correct decision.

4.4.2. Open quantum walk model (OQWM)

Attal et al. first introduced the quantum analogues of random Markov chains, namely open quantum walk, which is governed by the interaction with its surrounding dissipative environment in 2012. Further, a mathematical formulation for probability distribution, its quantum trajectory and time-dependent evolution of OQWM on graphs is discussed [130]. Additionally, Attal et al. [131] used the OQWM on complete connected graphs for dissipative state preparation, i.e. random single-qubit and two qubits Bell states and formulating quantum computing algorithms. Sinayskiy and Petruccione [132] provided a brief review on OQWM formulation on graphs as complete discrete positive maps. A summary of properties governing its evolution and its simulation using quantum trajectories is also provided. Optical implementation of the earlier introduced concept of OQW in [130] has been presented by Sinayskiy and Petruccione in [133]. In this implementation, a quantum walker is described using a two-level atom and lattice sites are described by Fock states of cavity mode. Additionally, they have described the asymptotic nature, probability distribution and properties of OQWM and its relation with the

generalized quantum coin spectrum [134]. Sinayskiy and Petruccione also presented the microscopic derivation of both discrete and continuous-time OQWM formulation on graphs by describing the relationship between OQW dynamics and its surrounding environment thermo-dynamical characteristics [135]. Pawela et al. [136] used transition operation matrices formalism to describe OQWM evolution on Apollonian networks to calculate average return time and mean initial passage times. Liu and Balu [137] introduced continuous-time OQWM as complete positive map formulation on graphs and proved that it always converges to a steady-state irrespective of the initial state of the system. Bardet et al. [138] discussed the passage times, the approximate number of visits a quantum walker makes on a particular node, recurrence concept, exit times, finite domain exit probabilities, and their application to calculate harmonic measures and solve Dirichlet problems with respect to irreducible OQWs. Furthermore, Sinayskiy and Petruccione [139] thoroughly reviewed the open quantum walk formulation on graphs and its collaboration with other existing quantum walk models such as continuous-time and discrete-time model.

4.4.3. Unitary and non-unitary walk model

Quantum walks following a unitary operation for evolution are said to be unitary quantum walks. Although, the unitary operator provides many benefits such as tracing of a walker, faster spread and mixing time of walk, but non-unitary evolution has been proved to optimize these properties of quantum walk and leads to a more uniform probability distribution and efficient resource utilization. A numerical validation by Kendon and Maloyer [140] approves that all correlations are eliminated at final measurement state in an optimal non-unitary walk of 't' steps using only $\mathcal{O}(\log(t))$ random bits. Liu [141] has provided the technique to transform any open quantum walk into its corresponding unitary walk evolution on finite graphs and lattices. Mochizuki and Obuse [142] numerically derived non-unitary quantum walks with real quasi-energy despite broken PT -symmetry (a combined time-reversal and parity symmetry). Kawasaki et al. [143] extended the work in [142] by constructing a three time-step non-unitary quantum walk to facilitate the study of multiple edge states stability and topological phases for PT -symmetry in open non-Hermitian quantum systems. Carvalho et al. [144] investigated the site recurrence problem of open DTQW and unitary coined quantum walk, whereby the time taken by the walker to return to its neighboring nodes is calculated, and it is found to be closely dependent on an additive interference parameter. Wang et al. [145] theoretically and experimentally realized momentum-time skyrmions in PT -symmetric quench dynamics (non-unitary) using single-photon DTQW which shares a major role in studying dynamic topological phenomena.

4.4.4. Inhomogeneous quantum walk model

Linden and Sharam [146] introduced inhomogeneous quantum walks, where the transition of a walker from one node to other depends not only on coin register but also on the position register state. They further classified such walks into periodic (position dependent) bounded and un-bounded time quantum walks for analyzing the asymptotic speed of walk. Konno investigated the dynamics of one-dimensional discrete-time space-inhomogeneous quantum walks. A path counting approach is used to prove localization behavior of the aforementioned walk [147]. Endo and Konno extended the work in [147] by deriving an expression for stationary parameter with an exponential decay with respect to position of the walker. This expression follows symmetry around the origin and guarantees localization based on initial coin state [148]. Furthermore, Konno et al. expanded their earlier work as described above by analyzing

three different limit measures, namely weak limit, stationary and time-averaged limit corresponding to spatial perturbations in the inhomogeneous walk dynamics. A universality class of quantum walks relating to weak limit measures is also proposed, which additionally contains even quantum walks with one defect [149]. Shikano and Katsura [150] numerically analyzed the eigenvalues of inhomogeneous evolution operator (coin operation depends on its position also) which proved the fractal nature of these quantum walks. Buarque and Dias [151] have investigated the energy spectra, localization properties and space-coin entanglement for inhomogeneous aperiodic DTQWs. Aperiodicity in these quantum walks come from space dependent coin operator where its distribution is neither random as in the case of Anderson localization nor periodic as Bloch-like.

4.4.5. Aperiodic quantum walk model (AQWM)

Ribeiro et al. [152] defined a novel quantum walk model and experimentally realized it using different kinds of aperiodic quantum coins ordered in aperiodic sequences, which leads to a wide variety of viable wave-function evolution. Later, Gullo et al. in their work in [153] have investigated the impact of aperiodic sequences, namely Thue–Morse, Rudin-Shapiro and Fibonacci sequence in energy-spectra and dynamics of a single quantum walker in discrete-time and space which can further be used for experimental realization of AQWM using optical elements. Recently, Pires and Queirós [154] have used the aforementioned aperiodic sequences in [153] to examine a class of DTQWs on a chain lattice, wherein displacement of the quantum walker is governed by binary aperiodic jumps.

4.4.6. Lackadaisical quantum walk model (LQWM)

The first lackadaisical (or lazy) quantum walk model was introduced in [56], which used a three-state coin allowing the walker to stay put with non-zero amplitude. Childs [116] mentioned the term lazy quantum walk in 2010, while showing the correspondence between DTQW and CTQW models. A formal description of quantum counterparts of lazy random walk, i.e. lackadaisical quantum walk was introduced in 2015 by Wong [155], wherein a Grover search for the marked vertex in a complete graph is simulated using LQWM with each vertex having 'k' self-loops. Addition of self-loops has been found to increase the success probability from half to unity in the Grover search described above. Further, Wong [156] proved LQWM to be faster in searching a marked vertex in a two-dimensional grid of N vertices than its loop-less counterparts. Numerical simulation of the Wong's model achieved an improvement of $\mathcal{O}(\sqrt{\log(N)})$ in time complexity than earlier running time of $\mathcal{O}(\sqrt{N\log(N)})$ in a total of $\mathcal{O}(\sqrt{N\log(N)})$ time-steps of the walk. In another work on spatial search by LQWM, Rhodes and Wong [157] proved that lackadaisical walks perform better with distinct weights on self-loops. The model has been validated for searching multiple marked vertices (k) on a complete bipartite graph, where success probability is improved if k is confined in the same partite sub-graph. Furthermore, Rhodes and Wong [158] derived a numerical expression, i.e. $Weight_{optimal} = \frac{Degree(Loopless_graph)}{total_nodes}$ for optimal weights adjusted on self-loops in LQWM for efficient spatial search on vertex-transitive lattices.

Giri and Vladimir [159] analyzed quantum dynamics and speed-up in spatial search by LQWM on one and two-dimensional lattices for certain values of weights on self-loops. Numerical simulations achieved a success probability of $\mathcal{O}(1)$ and $\mathcal{O}(\sqrt{\frac{t}{k} \log \frac{t}{k}})$ after 't' time-steps and for 'k' marked states in-case of one and two-dimensional lattice respectively. Wang et al. [160] provided an analytical formula for validating ballistic spread and localization feature of quantum walker in LQWM on one-dimensional lattice after infinite time-steps in the walk evolution. Nahimovs

investigated the dynamics of lackadaisical quantum walk evolution on a two-dimensional grid for searching a set of marked nodes (k). Additionally, a technique for efficient spatial search by adjusting the weights of self-loops to optimal values is also proposed [161]. Nahimovs extended his work in [161] by applying the earlier proposed LQWM on honeycomb and triangular grids. A speed-up in running time complexity is achieved from $\mathcal{O}(\sqrt{n\log(n)})$ to $\mathcal{O}(\sqrt{n\log(n)})$ for honeycomb and triangular grids by putting the weights of self-loops as $3/n$ and $6/n$ respectively [162].

4.4.7. Topological quantum walk model (TQWM)

Recently, topological quantum walks have emerged as a new quantum walk formulation owing to recent advancements in topological quantum materials. Wu et al. summarized the theoretical background and experimental simulations of TQWM along with its applications in machine learning to identify topological phases and phase transitions in quantum walks in [163]. Balu et al. [164] investigated the physical realization of TQWM on a circle with four vertices(n) using only two qubits($\log(n)$) and quantum gates count polynomial in n and independent of the number of time-steps. Simulating two-dimensional quantum walks poses many difficulties in terms of constraints and experimental resource availability. D'Errico et al. [165] provided a novel approach to photonic simulation of two-dimensional topological quantum walks, which uses a transverse wavefactor module of a single light radiation to encode quantum walkers position. Furthermore, Xie et al. in [166] presented the experimental realization of TQWM by simulating Su–Schrieffer–Heeger model which has been proved to be equivalent to TQWM. Time-averaged chiral displacements are used to measure the underlying topological invariants and localization effects in the dynamics of walk evolution.

4.5. Strengths and limitations of quantum walk models

Strengths and limitations of different DTQWM, CTQWM and other QW models have been illustrated in Tables 5, 6, and 7 respectively.

4.6. Coin operators

An essential component of the discrete-time quantum walk is the use of coin operators in the unitary evolution of the walk. A variety of coin operators have been studied and used in the existing literature. This section provides an account of the use of multiple coin operators in quantum walk dynamics, highlights the mathematical and analytical perspective of the coin operators and the effects that decoherence and entanglement between coin operators have on the dynamics of quantum walk.

4.6.1. Multiple coins in the unrestricted quantum walk

The impact of different and multiple coins on quantum walk dynamics has been studied by various authors [232]. A insightful article highlighting the impact of system's initial state and coin entanglement on the dynamics of coined quantum walk on graphs have been discussed by Tregenna et al. [232]. A mathematical perspective of various coin operators used for unitary operation during quantum walk evolution of graphs of degrees two, three, four and other higher dimensions was briefly discussed with its implementation on glued trees. Venegas-Andraca et al. [9] explored and analyzed the effect of entangled quantum coins on dynamics of coined quantum walk on a line. Further, a comparison between unitary evolution using a single coin, entangled quantum coins, and correlated classical coins is provided

Table 5
Strengths and limitations of discrete-time quantum walk model.

Application domain	Reference	Strengths	Limitations
Graph collision	Magniez et al. (2007) [167]	• small quantum memory required • query complexity optimizes to $\tilde{O}(n^{1/3})$ using recursion of dynamic version of Ambainis approach [4]	• Algorithm execution requires three different registers namely, set, data and coin register
Subset finding	Childs and Eisenberg (2003) [168]	• provides a concrete optimal lower bound of $\Omega(N^{1/L+1})$ for L-subset finding in database of N elements	• Algorithm does not perform well for graphs of higher degrees • continuous-time formulation offers more natural presentation
Graph Isomorphism	Douglas and Wang (2008) [169]	• difficult graphs, i.e. SRGs were optimally distinguished • GI algorithm utilized a novel measurement method	• approach does not apply to test general graphs • high run-time complexity $\mathcal{O}(n^7)$
Graph Isomorphism	Liu et al. (2019) [170]	• Polynomial time complexity • perfect accuracy for ordinary graphs • MapEff keeps bijections consistent with one another	• difficulty to work for regular graphs, i.e. graphs with inherent symmetry
Simulated Annealing	Somma et al. (2007) [171]	• speed up in number of operations • efficient resource utilization in simulation • proposed algorithm utilizes quantum Zeno effect and can be modified to simulate continuous-simulated annealing	• cubic scaling is worse than even classical simulated annealing
Universal Computation	Lovett et al. (2010) [118]	• gate constructs requires same number of wires as compared to CTQW • small overhead than continuous-time model • graphs having higher degrees used to ensure directional propagation	• gate constructs require twice the number of edges as compared to CTQW • phase gate requires an additional time step than continuous-time model
Graph connectivity, Connected components in graph	Montanaro (2005) [172]	• works for directed graph • provides partial quantum walk for irreversible directed graphs	• graph must follow reversibility condition • simulated only for one quantum walker
Graph connectivity, Connected components in graph	Albertini and Alessandro (2012) [173]	• provides controllability condition • gives dependency on topological properties of the underlying graph • coin operation changes at each vertex of graph	• proposed algorithm does not work for density matrices, i.e. unless ρ constitutes a pure quantum state
Detect marked vertices	Magniez et al. (2011) [88]	• quadratic speed up in hitting time over classical • reduced cost of finding marked vertex as compared to earlier models	• considers only single instance of marked element • applicable only on undirected graphs
Detect marked vertices	Krovi et al. (2010) [174]	• approach is simple and general • results remain valid for multiple marked vertices and when random walk P is not state-transitive	• additional complexity of vertex register maintenance • increased computational cost due to check, set-up and update black-box operations
Detect marked vertices	Hillery et al. (2010) [175]	• search quadratically faster than classical counterparts • works efficiently for small graphs	• search becomes difficult for complete sub-graphs with vertices $\geq$ four
Detect marked vertices	Tani (2008) [176]	• uses claw detection as a subroutine to claw finding algorithm • works for functions with different domain sizes	• query complexity increases $\log N$ times for k-function case
Detect marked vertices	Berry and Wang (2010) [177]	• polynomial time complexity of search • resultant success probability also provides information about topology of graph	• proposed partitioning approach not applicable to all graphs, specifically strongly regular graphs • considers only “closeness” as centrality measure for study
Spatial search	Potoček et al. (2009) [178]	• success probability close to unity $P_{\text{success}} \approx 1$ • reduced number of oracle queries and walk repetitions	• verification problematic for spatial search implementation where marked vertex is not oracle but a local property
Decision-making problems	Fuss and Navarro (2013) [179]	• decision model is based on cognitive and parallel architecture • includes decoherence and environmental noise while modeling quantum walk systems	• model works only for special case, i.e. optimal pattern of competitive and co-operative interactions
Steganography	EL-Latif et al. (2019) [180]	• proposed technique is efficient and has high security, visual quality, embedding capacity and performance • resistance towards quantum attacks	• increased complexity due to use of two coins and two walkers • complex embedding and extraction procedures

(continued on next page)

in [9]. The results obtained in [9] show similarities to those analyzed by Inui and Konno in [233], where a four-state coin is used to show localization effect in one-dimensional coined quantum walk around the origin of the walk. Moreover in [56], Konno

et al. derived an expression for the stationary distribution of a particle with three internal degrees of freedom, corresponding to moving in either left or right direction or resting in the same position. Lorz et al. [234] extended the coin degrees of freedom to

Table 5 (continued).

Application domain	Reference	Strengths	Limitations
Steganography	Abd-El-Atty et al. (2019) [181]	- high security and efficiency - large key space	- high key sensitivity - high complexity due to use of three unitary operators
Steganography	Yang et al. (2015) [182]	- highly secure, stable, unpredictable and non-periodic scheme with $\approx$ infinite key space - sensitive to system parameters and initial values	slower encryption speed as compared to existing schemes
Steganography	Yang et al. (2016) [183]	- privacy amplification using secure quantum key distribution - hash function acts as pseudo-random number generator and encryption algorithm	speed and randomness of proposed technique is limited due to classical simulation
Steganography	Peng et al. (2019) [184]	- highly secure technique - simple as only one channel is used to hide secret bits	only applicable to gray-scale secret images
Medical-image Steganography	Abd-El-Atty et al. (2020) [185]	- good visual quality, secure, increased embedding capacity and resistance to attacks - no requirement of pre and post encryption or extraction methods	- complex due to controlled alternate quantum walks - limited validation (ten images types only)
Internet-of-Things	Abd EL-Latif et al. (2019) [186]	- highly secure, efficient and robust - provides very large key-space	increased computational complexity due to controlled walks and different unitary operators
Security Protocol	Rohde et al. (2012) [187]	requires only one round of communication to establish a secure connection between client and server	no measurement or feed-forward within circuit is allowed
Cryptography protocol	Vlachou et al. (2015) [188]	- efficient and secure protocol - complexity scales polynomial in the size of message	requires access to quantum computer by both sender and receiver
Security Protocols	Chandrashekar and Busch (2015) [189]	- perfect recovery using temporal disorder in system - protocol secure against eavesdropping attack	- works only for systems with experimental control over walk dynamics - keeping information in quantum memory for a long time is difficult
Security Protocols	Abd EL-Latif et al. (2019) [190]	hash function and security protocols are highly efficient, secure, stable, large key-space and robust against various attacks	- designed specifically for 5G networks - complex due to different unitary operators
Secure hash function	Yang et al. (2019) [191]	- simple implementation and high security against forgery - effective collision resistance	- less efficient than SHA3, SHA256 hash functions - no mathematical validation of security
Teleportation	Wang et al. (2017) [192]	- no requirement for prior entangled state as it is created during first step of quantum walk itself - allows to teleport a d-dimensional state	- quantum particles are not co-located - applicable only on a certain family of graphs
Teleportation	Chatterjee et al. (2019) [193]	- teleportation done with high-constancy for one-qubit state - applicable to two and three qubit state	only mathematical description provided, no experimental validation
Teleportation	Li et al. (2019) [194]	- no explicit entangled state required as multiple walkers create one themselves - simplified experimental realization and flexibility of scheme	two coins are used and for success probability to be unity, the second coin must be unbiased
Teleportation and Quantum Communication Protocols	Shang et al. (2019) [195]	- perfect state transfer - external control over coin operator is minimal	validated experimentally only for line and circle
Quantum Communication Protocols	Yang et al. (2018) [196]	- success probability and state fidelity is unity for single qubit states - spatial resource constant	applicable only to inverted crown and butterfly networks
Machine learning and Evolutionary algorithms	Lovett et al. (2013) [197]	- efficient and adaptive metrology policies with run-time speed-up of $\approx 2^7$ - inclusion of feedback control and many-particle dynamics	- experimental evaluation of phase estimation for binary decision tree model only - does not consider factors such as decoherence and phase noise of multi-photon state
Algorithm design	Broome et al. (2010) [198]	- robust, scalable stable and deterministic approach - includes tunable decoherence impact on system dynamics	- only single coin operator and photon walk considered - simulated only six steps of one-dimensional walk
Quantum Neural Networks	Dernbach et al. (2018) [199]	Quantum walk neural networks are able to adapt both spatially and temporally over graph and walk respectively	- complexity may vary for isomorphic graphs - slow computation

(continued on next page)

four in their work to realize multiple ballistic propagation speeds and captured the dynamics of coined quantum walk on complex graphs.

Xue and Sanders [235] studied and analyzed the effects of partial and fully swapping of coins after each time-step among two independent quantum walks on line in terms of quantum mutual information and, measurement-induced disturbance. Yang

Table 5 (continued).

Application domain	Reference	Strengths	Limitations
Genetic Algorithms	Pitchai et al. (2016) [200]	• fuzzy attributes considered for modeling system • flexible requirement specifications • effective and practical algorithm	• team size representation is crisp but does not consider crisp constraints, i.e. budget limits
Genetic Algorithms	Koppaka and Hota (2010) [201]	• higher accuracy • faster convergence speed	• only 0-1 Knapsack problem considered
Algorithm design	Szegedy (2004) [6]	• faster search even for multiple marked vertices • quantum escape time depends on spectral properties of transition matrix	• works successfully only for a set of graphs such as bipartite graphs
Forecasting using machine learning	Konno (2019) [202]	• approach incorporates non-stationary time series	• applicable only to two-state one-dimensional system
Blockchain security	Chiang et al. (2020) [203]	• secure, polyvalent, efficient and client-centric system • highly random keys using pseudo-random number generator • avoids double spending problem	• increased complexity as it uses different protocols such as MALE (multi attribute lock encryption) and QKD (quantum key distribution) for encryption and key distribution
Metrology	Kitagawa et al. (2012) [204]	• time-dependent variation in bound state properties • robust to system perturbations	• simulated one-dimensional quantum walk only

Table 6

Strengths and limitations of continuous-time quantum walk model.

Application domain	Reference	Strengths	Limitations
Element distinctness	Ambainis (2007) [4]	• low complexity • generalized approach	• allows only comparison queries • no simulation, only mathematical validation
Hidden sub-graph problem	Zhang et al. (2019) [205]	• captures both global and local connectivity structure of graph • highly efficient and accurate for graph and node classification	• works well for a selected family of graphs and trees • added complexity of graph transformation function
Graph Isomorphism	Gamble et al. (2010) [206]	• effective in distinguishing strongly regular graph pairs of up-to 64 vertices • interacting particles powerful in distinguishing abilities than single or independent ones	• numerical validation only • proposed algorithm may break if GI(graph isomorphism) is not in P-class • two non-interacting walkers unable to identify GI in SRGs
Graph Isomorphism	Rudinger et al. (2012) [207]	• distinguishing power for non-isomorphic Strongly Regular Graphs (SRGs) increases with increase in the number of quantum walkers	• polynomial time algorithm does not exist for GI problem in SRGs • some pairs of SRGs cannot be distinguished by any two, three or m-particle non-interacting walk
Algorithms	Farhi et al. (2007) [208]	• works in Hamiltonian oracle which is an improvement over quantum query model • optimal complexity for NAND-tree evaluation	• model applicable only for NAND-tree formulations • works for only time-independent Hamiltonian
Universal Computation	Childs (2009) [117]	• efficient implementation of CNOT gate and single qubit gates • proves universal computation of quantum walks	• applicable only to sparse graphs, i.e. with degree equal to at-most three • no dynamic control of the system
Universal Computation	Childs et al. (2013) [209]	• polynomial size graphs suffice simulation of multi-particle walks with a very small error term • both wave packets and graphs finite • does not require time-dependent control for constructing a scalable quantum system	• algorithm is not fault tolerant • the bounds on vertex size, evolution time and size of the wave packets are not optimal
Universal Computation	Peruzzo et al. (2010)[210]	• powerful method for encoding information in exponentially large state space and hence scalable	• correlations among walkers critically dependent on initial state of system • simulated for lower complexity graphs only
Detect marked vertices	Chakraborty et al. (2016) [211]	• algorithm works well for disordered systems • inherent robustness to edge loss • high fidelity communication	• optimal search results only for value of probability 'p' that an edge exist in a graph of 'n' nodes equal to $p \geq \log^{3/2}(n)/n$
Detect hidden non-linear structures in graphs	Childs et al. (2007) [212]	• efficient query complexity	• algorithm works only for odd dimensions

(continued on next page)

et al. [236] extended the work in [235] by analyzing the walk dynamics where the movement of a quantum walker is controlled by two initially independent quantum coins alternately partial or full after each step. Karafyllidis and Haguel [237] used hybrid coins in quantum walk evolution on line where different quantum

coins were used for odd and even time-steps. This approach helped in quantum search as hybrid coins facilitated directional and confined search. Wang et al. [192] presented a generalized teleportation scheme for realizing communicating protocols using two quantum coins.

Table 6 (continued).

Application domain	Reference	Strengths	Limitations
Spatial search	Novo et al. (2015) [213]	• simplified computation • optimal search for star graph family • high fidelity on bound for single qubit transfer	• not optimal for highly symmetric graphs • bound is tight only when Hamiltonian is of spin network
Spatial search	Chakraborty et al. (2017) [214]	• optimal communication tasks and search on dynamical and temporal random networks • high fidelity in single qubit transfer through network	• search optimal for only single marked element • sub-optimal search for a temporal network which is composed of a sequence of static graphs
Graph partitioning	Carlson et al. (2006) [215]	• provides analysis of universal, instantaneous and average mixing times on edge-weighted graphs	• applicable only to undirected simple (no self-loops) graphs • not validated for structures of path length ≥ 4
Decision-making problems	Fuss and Navarro (2008) [216]	• very fast response time • different decision models designed by varying decoherence induced noise	• partial coherent walks face issues in response time modeling, explicitly in information gathering whether serially or parallel
Decision-making problems	Lloyd (2011) [217]	• optimal quantum transport at resultant temperature • optimum is robust	• success probability decreases if effective temperature crosses optimal mark
Decision-making problems	Busmeyer et al. (2006) [218]	• presented a simple way to model decision making for signal detection • choice probability and its response time included in model fit • high accuracy and	• assumes that model parameters such as initial start point and mean vary across trials • model does not consider effects of variability based on non-decision parameters such as motor response time
Machine learning and Optimization	Wossnig et al. (2018) [219]	• polynomial improvement in run-time complexity • efficient quantum state preparation	• large constant overhead w.r.t. fault tolerant gates • limited applications due to overhead
Similarity search in graph theory	Bai et al. (2013) [220]	• effective approach of classification • vast application in graph theoretic domain	• applied only on undirected graphs
Similarity search in graph theory	Bai et al. (2015) [221]	• polynomial time computational complexity and higher classification accuracy • works for directed graphs • solves missing vertex and permutation variance problems	• optimal for simple planar graphs, not applied on higher-dimensional graphs such as hyper-graphs • time-averaged density matrix computation required
Similarity search in graph theory	Rossi et al. (2015) [222]	• faster computation • efficient kernel • optimal classification accuracy at time T which is comparable in magnitude to average graph size in data-set	• eigenvalue decomposition overhead • applicable to unattributed graphs only • divergence minimum only for graphs with disjoint Hamiltonians
Algorithm design	Low and Chuang (2019) [223]	• query complexity is optimal • quadratic speed-up in space and gate complexity • considers both asymptotic and non-asymptotic schemes • low overhead	• eigen-decomposition difficult to compute • issues in Hamiltonian construction with large spectral norms • does not incorporate structural information

Rebeiro et al. in [152], introduced the concept of aperiodic quantum random walk wherein, several biased quantum coins are applied aperiodically during walk evolution on one-dimensional chain. Furthermore, Buarque and Dias [151] studied the energy spectrum, coin-position entanglement and, localization properties of aperiodic quantum walks using many spatially dependent quantum coins distributed on the lattice. Alessandro et al. [238] studied the behavior of non-stationary quantum walks on a cycle using a different coin operator at each computational time-step.

Linden and Sharam [146] provided a formal introduction to in-homogeneous quantum walk in which the coin operator depends not only on coin registers but also the position register. Shikano and Katsura have extended the work in [146] and studied the properties relating to fractality, self-duality and localization in in-homogeneous quantum walks and further proved that the limit distribution is localized at the origin of the walk [150]. Konno [147] presented and numerically proved a return probability theorem for an in-homogeneous periodic quantum walk with many coins and that the return probability can be represented in terms of elliptical integrals [239]. Moreover, Konno et al. [149] proposed a universality class of in-homogeneous walks with respect to weak limit measure for a walk starting at the origin. Other measures that were taken into consideration are time-averaged limit measure and stationary measure.

Brun et al. [8] analyzed the dynamics of coined quantum walk on a line using single coins of 2^M dimension and M two-dimensional coins and, Sewaga and Konno [240] provided mathematical and analytical expression for transition of quantum to classical walk based on three factors, i.e. the initial qubit, M number of coins and a factor $d = \lceil t/M \rceil$, where t denotes time steps. Schreiber et al. [241] realized a quantum walk evolution over five steps using only passive optical elements with the help of many different coin settings and initial system states.

Bañuls et al. [242] introduced the concept of time-dependent quantum coins in quantum walks exhibiting dynamical localization and, Machida and Konno [50] presented the limit distribution equations for such walks with $\hat{C} = \hat{C}(\text{time})$. Furthermore, Chandrashekar and Banerjee [243] presented a new variant of Parrondo game using two different coins which can be used either alternatively or in combination at each time step of the quantum walk evolution. Finally, Chung and Choon-Lin [244] presented a numerical analysis of localization and recurrence of a quantum walk model in periodic potential with two quantum coins.

4.6.2. Brief summary of widely used coin operators

An arbitrary three-parameter SU(2) operator [47] is the most commonly and widely used quantum coin operator in the quantum walk evolution on the line. It can be defined in its matrix form by the following equation

$$\hat{U}_{\xi, \beta, \zeta} = \begin{pmatrix} e^{i\xi} \cos(\beta) & e^{i\zeta} \sin(\beta) \\ e^{-i\zeta} \sin(\beta) & -e^{-i\xi} \cos(\beta) \end{pmatrix} \quad (27)$$

Table 7
Strengths and limitations of miscellaneous quantum walk models.

Quantum model	Application domain	Reference	Strengths	Limitations
QSWM	Graph Isomorphism	Bruderer and Plenio (2016) [224]	- optimal performance in partially coherent schemes - adaptable to decoherence	- simulated for undirected simple graphs only - difficulty in measuring cumulants with high precision
Szegedy quantum walk	Graph connectivity, Connected components in graph	Belovs and Reichardt (2012) [225]	- time-efficient algorithm - Optimal quantum query complexity of the order $\log(n)$ for a graph with n nodes	Query complexity to find a triangle in the given space is not optimal and is of the order $\mathcal{O}(n)$ for a graph with n nodes
Lackadaisical quantum walk	Graph connectivity, Connected components in graph	Wong (2015) [155]	- good search performance with single self-loop - success probability shifts from half to one after adding a self-loop on each vertex in DTQW	- more than one self loop deteriorates the performance - validated only for single marked vertex - no effect on CTQW
Staggered quantum walk	Detect marked vertices	Portugal (2016) [5]	- difference between coin and shift operator vanishes when coined quantum walk is converted to staggered walk (SQW), and they act as alternate unitary operators - SQW helps to set up an equivalence between coined and Szegedy's QW - SQW unifies DTQW models - SQW model may help to understand relation between coined and continuous quantum walk	- model validated on a certain family on graphs only, i.e. - line graphs of bipartite graph with and without perfect matching and non-bipartite graphs - not line graphs
Quantum random walk	Decision-making problems	Biamonte et al. (2019) [226]	efficient and enhanced quantum algorithms for complex network problems	no mathematical formulations or simulation
Quantum random walk	Decision-making problems	Sgarbas (2007) [227]	diverse application domains in artificial intelligence	no specific problem formulation, only theoretical description
QSWM	Machine Learning	Adcock et al. (2015) [228]	explores various machine learning domains such as quantum neural networks	no simulation or mathematical validation
QSWM	Quantum Neural Networks	Schuld et al. (2014) [229]	- moderate speed-up in complexity - optimized associative memory probability in case of two sinks at equal distance - performance enhanced by decoherence - low dependence on coherent part of walk	- efficient performance only under certain constraints - higher dimensional simulations $d \geq 7$ requires vast computational resources - quantum neurons not considered, only firing states of entire Quantum Neural Network (QNN) used - no dynamic learning mechanism
QSWM	Quantum transport in complex networks	Caruso (2014) [230]	- robust against geometry changes - decoherence and external noise included in walk evolution - works well for large complex network structures - efficient transport in network	- considers only connected graphs - work well only for simple graphs, i.e. no loops allowed - fails to give optimal transport efficiency for some family of well-connected graphs where multiple trapped sites arise due to destructive interference
QSWM	Quantum transport	Caruso et al. (2016) [231]	- transport efficiency increases with increase in maze size - experimental solution for hybrid approach - considers decoherence and noise-enhanced impact on system dynamics	- provides only theoretical solution for pure classical and quantum approaches - analyzed only regular and ordered graphs - difficulty in effectively mapping of maze into experimental setup

In Eq. (27) $0 \leq \beta \leq \pi/2$. It has been applied in unitary evolution of the discrete-time quantum walk in [47,245,246]. Replacing the value of $\beta = \pi/4$ gives the well-known Hadamard coin operator for quantum walk evolution also known as Hadamard walk on a line defined using Eq. (28)

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (28)$$

The Hadamard operator is well-known for its resultant ballistic evolution of quantum walk. Other variations of the quantum coin operator can be obtained by varying the values of β in the interval of 0 and $\pi/2$. For instance by putting the β value as $\pi/2$ and 0 gives us Pauli operators σ_x and σ_z represented as in Eq. (29)

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (29)$$

Another generic quantum coin which has found profound application in quantum walk evolution is defined as in Eq. (30)

$$\hat{C} = \begin{pmatrix} \cos \beta & i \sin \beta \\ i \sin \beta & \cos \beta \end{pmatrix} \quad (30)$$

This symmetric quantum coin and its derivatives such as following

$$\hat{C} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad (31)$$

and other variations have been extensively used in literature.

For graphs of higher dimensions (d -dimensional graphs), i.e. graphs with degree three or more, two different types of coins can be used. One is the Discrete-Fourier Transform coin (DFT coin) which is unbiased and balanced but asymmetric. The other coin is Grover's operator, which is symmetric but biased as the probability of selecting different paths varies with the direction. The mathematical formulation of both the types (DFT and Grover's)

are provided below in Eqs. (32) and (33), respectively.

$$Coin_{DFT} = \frac{1}{\sqrt{d}} \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \alpha & \alpha^2 & \dots & \alpha^{d-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{d-1} & \alpha^{2(d-1)} & \dots & \alpha^{(d-1)(d-1)} \end{pmatrix} \quad (32)$$

Here α is the d th root of unity, $\alpha = e^{\frac{2\pi i}{d}}$.

$$Coin_{(x,y)}^* = \begin{pmatrix} x & y & y & \dots & y \\ y & x & y & \dots & y \\ y & y & x & \dots & y \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y & y & y & \dots & x \end{pmatrix} \quad (33)$$

Here x and y are real and $1 - \frac{2}{d} \leq |x| \leq 1$, $y = \pm(1 - x)$. For Grover coin, x and y equals to $\frac{2}{d} - 1$ and $\frac{2}{d}$ respectively. Table 8 presents a summary of the different coin operators studied in our review.

4.6.3. Entanglement in quantum random walks

Panahiyan and Fritzsche [250] investigated the relationship between system entropy and initial entanglement in qubits arising due to internal degrees of freedom. The results showed that entropy is minimum in case of separable qubits whereas it is maximum if qubits are maximally entangled initially. Venegas-Andraca et al. numerically described asymptotical properties (specially the “three-peak zone”) of quantum walks for entangled coins in [9] and reported higher probability of finding the quantum walker at $|0\rangle$, i.e. origin ($p \approx 0.221622$) in quantum case than in classical ($p \approx 0.0795$). Later in [251], Venegas-Andraca proposed generalized coined quantum walks with varying degree of entanglement in coin pairs and initial states as quantum walkers in superposition. Buarque and Dias in [151] studied the impact of aperiodicity on position-coin entanglement in case of static in-homogeneous systems. They have analyzed that such systems display asymptotic limit of entanglement. A novel technique for measuring the fairness of a quantum coin and quantum states discrimination is devised using entanglement theory in [252].

Gratsea et al. [253] proposed two methods for generating highly entangle quantum states which have very little or no dependence on initial states of the system. In the first approach, a deterministic chain of coin operators is used while the second approach includes optimizing the coin sequence using a reinforcement learning technique. Apel et al. [254] investigated the entanglement properties of a particle-pointer system. An exact solution is provided for the Schrödinger equation governing the evolution of particle-pointer system. In a recent study, Gambhir and Singh [255] have experimentally simulated the diffraction-pattern of photons exhibiting position-momentum entanglement using a sharp edge. Bergschneider et al. [256] provided a technique to verify and assess the entanglement entropy of individual particles and used it to derive coherence and correlations among them. Chinese physicists have presented a method to fully entangle eighteen qubits by utilizing the position, polarization and orbital angular momentum of six photons which provides new insights into quantum information processing [257,258].

4.7. Existing simulators for quantum walk

This subsection provides a brief account of the various quantum simulators being used in the research in the quantum field. A comparative study of different existing quantum walk simulators in literature is presented in Table 9.

- (a) Hiperwalk: Lara et al. developed Hiperwalk, a program which simulates dynamic evolution of quantum walk models using hybrid High-Performance Computing (HPC) architectures [259]. It currently simulates discrete-time quantum walk and staggered walk and will be able to simulate Szegedy's walk and continuous-time quantum walk in the future. It is available in Python, Neblina, GNUplot and OpenCL languages and uses open-source non-proprietary codes. Major advantages include its user-friendly interface and use of hybrid HPC architecture for computation. The major limitation includes that it is in running only for discrete-time and staggered quantum walks.
- (b) pyCTQW: Izzac and Wang [260] developed a distributed-memory software framework, pyCTQW in python and FORTRAN programming language for propagation and visualization of large multi-particle (currently three walkers with interactions) continuous-time quantum walk on arbitrary graphs. The package uses Chebyshev and Krylov-subspace approximation methods for quantum walk propagation calculations.
- (c) qwViz: qwViz is a software package proposed by Berry et al. [261] to visualize discrete-time quantum walk on undirected complex arbitrary graphs. It is developed in C language and uses OpenGL to provide three-dimensional visualization of time-dependent probability distributions arising from DTQW on complex graphs. Input to the package can be given by either using the adjacency matrix corresponding to the graph or using Quantum Walk Markup Language (QWML).
- (d) QWalk: Marquezino and Portugal [262] developed C language simulator for discrete-time quantum walk on one and two-dimensional finite lattices of generic topology. It also includes simulation of decoherence by performing intermediate measurements or by disconnecting the links of the lattice under consideration. Qwalk provides two-shift operators for two-dimensional lattice and three-shift operators for a one-dimensional lattice. It can further be used to study walk with different boundary conditions and by appropriately defining perennial broken links in the lattice.
- (e) Quantum Computer Simulator (QCS) system: Sawerwain and Gielerek presented QWalk using HPC [263]. Simulations of standard 1D and 2D quantum walks have been performed within Quantum Computer Simulator (QCS system) environment and with the use of GPGPU (General Purpose Graphics Processor Unit) supported by CUDA (Compute Unified Device Architecture) technology.
- (f) QSWalk: Falloon et al. proposed a Mathematica package for the simulation of open continuous-time Quantum Stochastic Walk (QSW) on arbitrary directed graphs [123]. It takes into consideration the environment and the energy dissipation due to system bath interaction. Limitation includes issues with the performance of simulation of high dimensional systems and it is not applicable on large graphs due to memory size limitation. Only a single walker is considered during walk evolution using QSWalk. Major application areas are FMO complex in photosynthesis, quantum page rank algorithm, quantum-classical transition on a line and pure dephasing scattering among others.
- (g) QSWalk.jl: Glos et al. in [124] proposed a Julia framework that takes into consideration multi-particle system and only computes continuous-time stochastic evolution of quantum walks on arbitrary graphs.

Table 8
Description of coin operators.

Base quantum coin	Reference	Derived coin operator	Type of graph	Summary	N	Initial state of the system	Type of walk	Complexity speed-up or advantages	Application domain
SU(2)	Latif et al.(2019) [247]	$\hat{C}_0 = \begin{pmatrix} \cos \beta_0 & \sin \beta_0 \\ \sin \beta_0 & -\cos \beta_0 \end{pmatrix}$ and $\hat{C}_1 = \begin{pmatrix} \cos \beta_1 & \sin \beta_1 \\ \sin \beta_1 & -\cos \beta_1 \end{pmatrix}$	circle	two coin operators having values of β_0 and β_1 as $\pi/6$ and $\pi/3$ are used respectively	two	initial state depends on the i th bit of a string of 0 and 1 associated with each step of the walk	closed controlled alternate quantum walk	higher efficiency and security	privacy preservation in IoT based healthcare systems
SU(2)	Wu et al.(2019) [246]	$\hat{C}_{su} = \begin{pmatrix} e^{i\xi} \cos(\beta) & e^{i\xi} \sin(\beta) \\ -e^{-i\xi} \sin(\beta) & e^{-i\xi} \cos(\beta) \end{pmatrix}$ and $\hat{C}_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$	1-D infinite chain	$\hat{C}_{su}$ is the used basis SU(2) operator where ξ, β, ζ are real parameters and $\hat{C}_0$ is obtained by replacing the values of ξ, β, ζ as 0, $\pi/4$, and $\pi/2$ respectively	one	$\psi_0 = 0\rangle_p \otimes \frac{1}{\sqrt{2}}(0\rangle_c + 1\rangle_c)$	closed quantum walk and Classical Walk with Memory(CWM)	CWM shows better performance in terms of spreading properties of walk for few number of steps, while quantum walk is better if number of steps is large	optimizing complex algorithms, and data searching
SU(2)	Orthey and Amorim (2019)[245]	$\hat{C}(p, \alpha, \beta) = \begin{pmatrix} \sqrt{p} & \sqrt{1-p}e^{i\alpha} \\ \sqrt{1-p}e^{i\beta} & -\sqrt{p} \end{pmatrix}$	1-D graph	two unbiased coins ($p = 1/2$), i.e. Hadamard and Fourier coins having values of ($\alpha = \beta = 0$) and ($\alpha = \beta = \pi/2$) respectively are used to operate over spin states	one	$\psi_0 = \sum_{j=-\infty}^{j=\infty} [a(j, 0) \uparrow\rangle + b(j, 0) \downarrow\rangle] \otimes j\rangle$, $ j\rangle$ represents position vectors, and $a(j, 0)$ and $b(j, 0)$ corresponds to spin-up and down respectively	closed discrete quantum walk	average variance are 0.31 and <0.01 , and average entanglements after 3000 time-steps are 0.69 and 0.76 for Hadamard and Fourier coins respectively	build entanglement protocols and in quantum transport
SU(2)	Orthey and Amorim (2017) [40]	$\hat{C}(p, \alpha, \beta) = \begin{pmatrix} \sqrt{p} & \sqrt{1-p}e^{i\alpha} \\ \sqrt{1-p}e^{i\beta} & -\sqrt{p} \end{pmatrix}$	line	$\hat{C}(p, \alpha, \beta)$ operates over spin states and generates a superposition of them. The values of three independent parameters vary as, $0 \leq p \leq 1$, if $p = 1/2$ means unbiased coin, else if $p \neq 1/2$ means biased coin. $0 \leq \alpha, \beta \leq 2\pi$ controls the relative phases between spin states	one	any initial qubit with a spin-1/2	closed	average variance for a walk with local initial state does not depend on coin, whereas for Gaussian and uniform states it depends greatly on coin operators, being non-dispersive for $\alpha + \beta = \pi$ in case of Fourier walk	efficient quantum transport and protocol design

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Table 8 (continued).

Base quantum coin	Reference	Derived coin operator	Type of graph	Summary	N	Initial state of the system	Type of walk	Complexity speed-up or advantages	Application domain
Generic quantum coin	Kumar et al. [248]	$\hat{C}_\theta = \begin{pmatrix} \cos \theta & -i \sin \theta \\ -i \sin \theta & \cos \theta \end{pmatrix}$ and $\hat{C}_{\theta(j)} = \sum_j \hat{C}_{\theta(j)} \otimes j\rangle\langle j $	1-D lattice	$\hat{C}_{\theta(j)}$ is the position dependent coin operator $\hat{C}_\theta$ can be represented as a combination of Pauli matrices as $\hat{C}_{x,\theta} = e^{-i\theta \hat{n}_x \hat{\sigma}_x}$, where $\hat{n}_x$ is the unit vector	one	$ \Psi\rangle_{in} = [\cos(\delta) 0\rangle + e^{i\theta} \sin(\delta) 1\rangle] \otimes j=0\rangle$, denotes two state(coin) particle and positioned at origin	open discrete walk	temporal disorder enhances entanglement whereas spatial disorder decreases it	metrological domain to exploit dynamics of many-body systems
Hadamard	Venegas-Andraca et al.(2005) [9]	$\hat{C}_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	un-restricted line	probability distribution of maximally entangled Hadamard coins is observed with different initial states	one	$ \psi_0\rangle = pos\rangle_0 \otimes coin_0^*\rangle$, where $ coin_0^*\rangle = Bellstates X\rangle^+$, $ X\rangle^-, \eta\rangle^+, X\rangle^+ = 1/\sqrt{2}(00\rangle + 11\rangle)$, $ X\rangle^- = 1/\sqrt{2}(00\rangle - 11\rangle)$, $ \eta\rangle^+ = 1/\sqrt{2}(01\rangle + 10\rangle)$	closed entangled walk	three-peak zone achieved in probability distribution and symmetrical probability distribution achieved using entanglement	quantum computation with multiple coins
Generic quantum coin	Chandrashekar and Busch (2014) [249]	$\hat{C}_x = \begin{pmatrix} \cos \theta & -i \sin \theta \\ -i \sin \theta & \cos \theta \end{pmatrix}$ and $\hat{C}_y = \begin{pmatrix} \gamma & \eta \\ \eta & -\gamma \end{pmatrix}$	square lattice	directed DTQW evolution on completely connected square lattice of degree r and r-1 self-loops at each vertex, $\hat{C}_y$ values of γ and η are $\frac{1}{\sqrt{r}}$ and $\sqrt{\frac{r-1}{r}}$ respectively	one	$(i) \uparrow\rangle, (ii) \downarrow\rangle$, and $\frac{1}{\sqrt{2}}(\downarrow\rangle + i \uparrow\rangle)$ with $\theta = \frac{\pi}{4}$, $r = 2$	directed discrete-time quantum walk	system allows variation in quantum storage applications and efficient transport by varying percolation threshold	quantum transport
Grover coin	Yang et. al (2016) [183]	$\hat{C}_a = \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$ and $\hat{C}_b = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{pmatrix}$	circle	$\hat{C}_a^{16}$ is the Grover operator and $\hat{C}_b^{17}$ is the coin operator, the kth bit of the message controls the associated kth step of the quantum walk	two	$ x, y\rangle = u 00\rangle + v 01\rangle + w 10\rangle + z 11\rangle$	discrete-time quantum walk	very high key size $\approx 2^{325}$ results in protection against majority brute force attacks	Encryption algorithms design,pseudo-random number generation

N: Number of simultaneous quantum walkers in the system.

Table 9

Comparative study of various quantum walk simulators.

Reference	Name of the simulator	Programming language	Operating system	Type of quantum walk	QS type	N
Lara et al. (2015) [259]	Hiperwalk	Python, Neblina, Gnuplot, and OpenCL	linux, windows	DTQW	C	S
Izaac and Wang (2015) [260]	pyCTQW	Fortran, Python	NR	CTQW	C	M
Berry et al. (2011) [261]	qwViz	C language	linux, mac OS X 10.5 (and later)	DTQW	C	S
Marquezino and Portugal (2008) [262]	QWalk	C language	linux, windows	DTQW	C	S
Sawerwain and Gielerek (2010) [263]	QCS system	C language	OS with CUDA	DTQW, and CTQW	C	NR
Falloon et al. (2017) [123]	QSWalk	Mathematica	NR	quantum stochastic walk	O	S
Glos et al. (2019) [124]	QSWalk.JL	Julia	NR	quantum stochastic walk	O	S

QS Type: Type of quantum system (Open (O) or Closed(C)).

N: Number of simultaneous quantum walkers in the system (Single (S) or Multiple (M)).

NR: No restriction.

4.8. Quantum walk in physical systems

There have been various proposals for physical realization of different types of quantum walks utilizing a diverse array of optical schemes as well as solid-state. In this review, we present a systematic survey of the various implementation approaches and methods thus far proposed, and in some cases experimentally implemented, grouped according to the physical medium they essentially feature in Table 10.

Although, a number of authors have considered the implementation of CTQWs, a vast majority of the existing and proposed systems up-to date focus on the realization of DTQWs or coined quantum walks, thus requiring the elementary physical system to exhibit two different types of states, the addressable states which represent position states or nodes as well as the coin states or sub-nodes. It allows selective interactions between these states to implement a suitable coin operator, and also allows some type of state-dependent time evolution to implement the shift operator or conditional translation [264,265].

4.9. Decoherence and noise handling in quantum walk

Decoherence is a phenomenon that occurs when a quantum system interacts with its surrounding environment. Decoherence is generally modeled as a non-unitary evolution of quantum walk, so it requires extra formalism for representing mixed states using density matrix operators. Until recently, decoherence was assumed to be an annoyance as it was closely associated with loss of information in the quantum system. However, it has been explored that decoherence phenomena can indeed be beneficial in formulating the physical and natural occurring processes in the environment [210,307–309] and also produce interesting outcomes in quantum computation and information processing tasks [8,55,224,310].

Decoherence plays a key role in understanding the transition between classical and quantum versions of the random walk. Disorders can be introduced in quantum walk architecture mainly via position space of quantum walker and on quantum coin [23]. Disorder is also introduced due to system measurement or interaction of the quantum system with its surroundings [311]. This research domain helps in understanding the computational properties of quantum walks and may also help the aspiring researchers to provide some threshold or benchmark criteria to check the quantumness of quantum computers and also pave the way for developing quantum algorithms. Another important aspect of decoherence lies in simulating classical random walks using their quantum counterparts [312–315]. This will provide a better understanding of the dynamics relating to both types of walk and also, about the laws governing computational

processes in physical systems. Thus, for acquiring a detailed understanding of the mathematical and physical relations between quantum systems and their surrounding environment, along with exploring new paradigms for developing quantum algorithms, analyzing decoherence properties and its impact on quantum walk dynamics is an essential part of quantum computation.

Chandrashekar et al. [316] explored the impact of various noise types namely, bit flip, phase flip and amplitude damping channels on the dynamics of biased and unbiased Hadamard walk on a line and cycle. Annabestani et al. [317] generalized the technique proposed earlier by Brun et al. [8] for discrete-time quantum random walk evolution in the presence of decoherent quantum coin by providing mathematical and analytical expressions for various kinds of decoherence. Further, Annabestani et al. investigated the impact of shifting position decoherence, that exists due to the tunneling effect in the experimental implementation of one-dimensional discrete quantum walk [318]. A detailed account of the impact of spatial and temporal disorders on the dynamics of one-dimensional quantum walks has been proved mathematically and analytically in [248,311].

Several researchers have addressed the computational and physical properties of decoherence and disorders in quantum walks. Ermann et al. [319] explored the variations in quantum walk properties due to the presence of decoherence induced by regular and chaotic environments using a complex coin. Further, Jayakody and Nanayakkara [62] provided a spin decoherence model based on quantum coin internal degrees of freedom, where bit-flip and phase-flip operators are applied randomly on the coin state. Chen and Zhang [320] explored the impact on position distribution of a two-dimensional walker in the presence of a coin decoherence model and broken line noise channel in an alternate quantum walk. Nejadstattari et al. [308] demonstrated the localization and revival of the wave function in photonic realization of cyclic quantum walks in noisy environments (bit-phase flip channel). Some authors have used Floquet operators to explore the impact of dynamic quantum phase transitions on coined quantum walks [321,322]. Decoherence is further used for experimental detection of topological variants in non-unitary coined quantum walks and revealing the phase transition in photonic walks [323,324].

Schreiber et al. investigated the impact of static disorders and decoherence in photonic implementation of the one-dimensional coined quantum walk. They used controlled time-dependent parameters to simulate different environmental noise in the quantum system resulting in ballistic, diffusive and Anderson localization effects in coined quantum walk architecture. Brun et al. [8] have shown that quantum to classical transition can be done in two ways alongside performing state measurements, i.e. using decoherence in coin (internal degree of freedom) and by higher-dimensional quantum coins. In addition, many authors [224,325–

Table 10

Physical implementation of quantum walks.

Reference	Method/Technology/Approach	Type of walk	Lattice type	Application areas	D	N
Knight et al. (2003) [266]	optical waves and cavity using polarization cebits	coined quantum walk	line and circle	implementing higher dimensional quantum walk	✗	single
Roldàn and Soriano (2005) [267]	optical cavity	2-D coined QW	line	quantum computation	✗	single
Knight et al. (2004) [268]	long wavelength approximation	coined quantum walk	line	quantum gas lattice automata	✗	single
Knight et al. (2003) [269]	light wave in a dispersible optical fiber	two independent coined QWs	line	–	✗	single
Jeong et al. (2004) [270]	beam-splitters, phase-shifters and photo-detectors	coined quantum random walk	line	design quantum algorithms for simulation walks on higher dimensional graphs	✓	single
Rai et al. (2008) [271]	array of coupled waveguides	CTQW	one-dimensional lattice	quantum transport	✗	multiple photons
Schreiber et al. (2010) [241]	passive optical elements	coined quantum walk	line	quantum search algorithms	✗	mutple photons $ H\rangle$ and $ V\rangle$ polarized photons
Zhang et al. (2010) [272]	optical device “q-plate”	DTQW	line	simulating quantum random walk algorithm on a quantum computer	✗	multiple photons
Rohde et al. (2011) [76]	quantum optics and photonic implementation	DTQW	arbitrary graph structures	quantum computation	✗	multiple walkers two and multiple photons
Solntsev et al. (2012) [273]	Spontaneous parametric down-conversion	continuous-time simultaneous quantum walk	quadratic nonlinear waveguide array	developing quantum integrated circuits, combining photon-pair generation and correlated photon-states transformation	✗	photon pairs
Witthaut (2010) [274]	ultra-cold spinor atoms	discrete-time Bloch walk	tilted and driven optical lattices	quantum information and simulation	✓	single
Hoogdaem and Blaauboer (2009) [275]	Zeeman splitting and electron spin	DTQW	linear chain of quantum dots	quantum transport	✓	single electron
Ribeiro et al. (2008) [276]	Parametric down conversion, single-photon	discrete quantum random walk	one-dimensional lattice using Sagnac interferometer and bi-refringent crystal	–	✗	single photon
Ezawa (2011) [277]	Skyrmion burst, thin ferromagnetic films	CTQW	two dimensional lattice	investigate dynamics of quantum walk	✗	two, multiple
Owens et al. (2011) [278]	Photons using Spontaneous Parametric Down Conversion (SPDC)	CTQW	3-D waveguide structure(elliptical direct-write waveguide array)	quantum emulation in quantum walks	✗	two photons
Oka et al. (2005) [279]	electron systems and Landau–Zener transition dynamics	CTQW	semi-infinite one-dimensional lattice having a reflecting boundary	explore non-equilibrium dynamics of driven quantum systems	✓	–
Hamilton et al. (2011) [280]	Photonic walk using Orbital Angular Momentum (OAM)	one and two-dimensional DTQWs	one-dimensional lattice	quantum search algorithms	✗	single photon
Kålmàn et al. (2009) [281]	electron-spin and spintronic semiconductor quantum rings	coined quantum walk	one-dimensional lattice	quantum transport and computing	✗	single electron

(continued on next page)

Table 10 (continued).

Reference	Method/Technology/Approach	Type of walk	Lattice type	Application areas	D	N
Du et al. (2003) [282]	two-qubit Nuclear Magnetic Resonance (NMR) quantum computer	CTQW	circle with four nodes	develop quantum random walk algorithms	✗	single
Schreiber et al. (2012) [283]	optical-fiber network	coined quantum walk	non-trivial graph structure	simulating quantum information, transport system and quantum search algorithm	✗	two walkers
Jeong et al. (2013) [284]	SPDC and type-II PPKTE crystal	DTQW and Grover-walk	two-dimensional lattice	Grover quantum-search algorithm	✗	single photon
Melnikov and Fedichkin [285]	array of tunnel-coupled semiconductor quantum dots	CTQW	cycle	quantum information processing and computation	✗	two electrons interacting and non-interacting
Apers et al. (2018) [286]	Lifted Markov Chains (LMC)	coined quantum walk	two-dimensional graphs	sampling and Monte-Carlo simulations	✓	single
Tang et al. (2018) [111]	quantum photonic chips	CTQW	hexagonal graphs of upto 160 nodes and eight-layers of depth and modified glued trees	quantum simulation, optimization and management tasks, and information processing	✗	single photon
Tang et al. (2019) [128]	three-dimensional quantum photonic chip	QSWM	one-dimensional array of seven waveguides	simulating associative memory in Hopfield neural networks, designing photonic AI devices for optimization computing	✓	single
Dadras et al. (2019) [287]	optical lattice of ultra-cold rubidium-87 atoms	DTQW	one-dimensional graph	engineering asymmetric quantum walks and atom interferometry	✓	single
Lorz et al. (2019) [234]	time-multiplexing in a Michelson-interferometer loop architecture	DTQW	line, circles of different size and Husimi cactus graph	quantum Grover walk, search algorithms and graph isomorphism problem	✓	single
Chatterjee et al. (2019) [288]	five-qubit quantum computer and 32-qubit simulator	coined quantum walk	one and two-dimensional graphs	quantum teleportation	✗	single
Caruso et al. (2016) [231]	integrated three-dimensional waveguide array	QSWM	maze-like two-dimensional graph	developing bio-inspired photonic technologies	✓	single walker
Broome et al. (2010) [198]	optical elements and photon	DTQW	a line with absorbing boundaries	energy transport and quantum computation	✓	single photon
Nejadsattari et al. (2019) [289]	Photonic simulation, quantum lattices	DTQW	one and two-dimensional lattices	quantum computation and communication systems	✗	single

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Table 10 (continued).

Reference	Method/Technology/Approach	Type of walk	Lattice type	Application areas	D	N
Xue and Sanders (2008) [290]	Quantum quincunx (QQ) variant in quantum cavity electrodynamics	coined quantum walk	one-dimensional lattice	quantum computation and design of algorithms	✗	single walker
Manouchehri and Wang (2009) [291]	Bose–Einstein condensate trapped in an optical lattice	DTQW	general undirected graph	designing quantum algorithms for efficient graph traversal	✗	single
Karski et al. (2009) [292]	spin-dependent optical lattice	DTQW	line	quantum cellular automata, quantum information science	✓	single
Schmitz et al. (2009) [293]	linear ion trap	Asymmetric coined quantum walk	line	laser systems	✗	single ion
Xue et al. (2009) [59]	Ion-trap lattice	coined quantum walk	line	designing quantum information protocols	✓	single ion
Xue et al. (2008) [294]	quantum quincunx via circuit QED (quantum electrodynamics)	DTQW	circle	–	✓	single walker
Yan et al. (2019) [295]	quantum processor and super-conducting qubits	CTQW	one-dimensional array	simulating many-body systems and universal quantum computation	✓	one and two photons
Giordani et al. (2019) [296]	linear-optics and OAM	DTQW	one-dimensional lattice	high-dimensional quantum state engineering	✗	single walker
Lavička et al. (2011) [297]	optical multi-ports	DTQW	one-dimensional	simulating complex system and game theory	✓	single walker
Zähringer et al. (2010) [298]	Ion-trap implementation	CTQW	line	quantum optics experiments and simulation	✗	one and two ions
Manouchehri and Wang (2008) [299]	quantum dot grid	DTQW	arbitrary complex two-dimensional graphs	algorithms for efficient graph connectivity and graph theory	✗	single electron
Ahlbrecht et al. (2011) [300]	optical lattice and Cs atoms	DTQW (free and interacting quantum walks)	one-dimensional graph	–	✗	two atoms
Mayer et al. (2011) [301]	unbiased beam-splitters	DTQW	one-dimensional	simulating many-particle quantum transport over disordered medium	✗	many particles
Matsuoka et al. (2011) [302]	optical frequency comb and diatomic molecule	CTQW	one-dimensional	simulating diffusion in quantum systems	✓	–
Lahini et al. (2012) [303]	non-linear photonic lattices	CTQW	–	simulating many-body systems	✗	two interacting bosons
Peruzzo et al. (2010) [210]	waveguide arrays in a SiOxNy (silicon oxynitride) chip	CTQW	two-dimensional lattice	quantum information encoding	✓	two identical photons
Sansoni et al. (2012) [304]	integrated three-dimensional optical circuit	DTQW	one-dimensional	simulation of interruption in periodic quantum lattice	✗	two non-interacting bosons, fermions
Regensburger et al. (2011) [305]	coupled fiber loops	DTQW	one-dimensional	design quantum algorithms and multi-pulsing fiber ring lasers	✓	single particle
Rohde et al. (2011) [306]	linear optics	DTQW	arbitrary graphs	universal computation	✗	multi photons

N: Number of simultaneous quantum walkers in the system.

D: Decoherence present in the system.

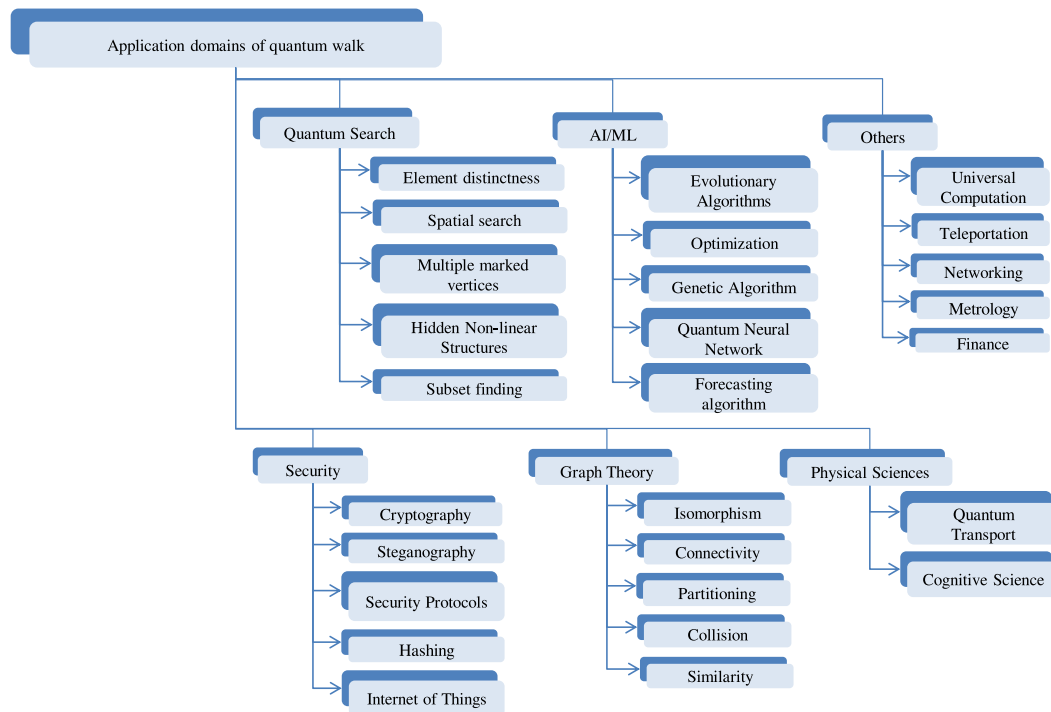


Fig. 14. Application domains of quantum walk.

327] have studied the decoherence properties in quantum walk evolution on a higher-dimensional lattice. A detailed account of the recent work relating to the study of decoherence and disorders in quantum walk dynamics is presented in Table 11.

5. Application domains of quantum walk

In this section, numerous emerging application areas of quantum walks have been underlined and discussed. These application domains have then been categorized into sub-domains for a better comprehensive analysis. Fig. 14 represents the categorization of various application domains and sub-domains of quantum walk model.

5.1. Broad classification of application domains of quantum walk

Quantum walk has emerged as a universal computational framework for a plethora of application domains, the majority of which lies in designing and validating quantum walk algorithms [17]. Quantum mechanics principles such as superposition, interference and others result in many exemplary attributes of quantum walks owing to the exponential and quadratic spread of the walk over sample space. This speed-up in their spreading behavior has made quantum walks an appropriate choice to be used in designing algorithms and quantum simulations of classical algorithms. Additionally, quantum computers have also shown to achieve quadratic speed-ups in mixing times while implementing Markov chains with a quantum walk [329].

Quantum walks have been used as a modeling tool in various domains such as quantum information processing, designing of quantum analogues of classical algorithms for hard problems, quantum computation, quantum automata design etc. Quantum walk provides exponential algorithmic speed up in terms of graph traversal metrics such as hitting, commute and cover times [2]. Many hard problems have been solved in polynomial time using quantum walks such as triangle problem, also known as graph collision problem [167], element distinctness problem [4], subset

finding problem [168], Graph Isomorphism (GI) [169,170,206,207,224,265], constraint satisfaction problem [330] and hidden sub-graph problems [205]. Further, quantum walks can be used as a tool for Universal Computation (UC) [117,118,209,210,265], the evolution of quantum algorithm for binary NAND tree in Hamiltonian oracle [208] and annealing processes [171]. Quantum walk has also been used for finding Graph Connectivity (GCn) in undirected and directed graphs, i.e. finding the number of connected components in graph [172,173,225], detecting or finding marked vertices [5,88,155,174–177,211] and hidden non-linear structures in a graph [212]. Quantum walk forms the core of many quantum algorithms for Quantum Search (QS) and spatial search [178,211,213,214,223], unstructured or slightly structured search problems [6,11,155,198] and graph partitioning [5,215].

Recently quantum physics principles have been applied to the human brain to explain human consciousness and automate the solution to planning and decision-making problems [179,216–218,226,227]. Also, quantum computing has been widely evolved in financial services for stock market prediction etc. [331]. Discovery of cryptocurrency for financial transaction in blockchain framework [203] has necessitated the requirement of secure key encryption, data hiding and secure key distribution and steganography [180–185]. Further, Quantum Security (Sec) finds application in smart cities, homes etc. using Internet of Things and its data security by designing efficient protocols [186–191] and other quantum communication protocols (QNC) [195,196]. Teleportation (TP) in quantum systems, i.e. transfer of energy from one qubit to some other state or Perfect State Transfer (PST) has been applied in the design of various algorithms and to realize the physical implementation of quantum walks [192–194].

Artificial intelligence and Quantum Machine learning (AI/ML) have recently emerged as one of the powerful domains as quantum walks provide increased computational power and speed to design machine learning algorithms [228,329,332,333], quantum information processing and evolutionary algorithms [197], optimization [219,334–337], genetic algorithms [200,201,338,339], QNN (quantum neural networks) in which a neuron is replaced by a qubit [199,229,340], and forecasting [202]. Some other widely

Table 11

Brief study of recent work on decoherence and noise handling in quantum walks.

Reference	Quantum walk type	Disorder type	Applications domain	Graphs used	N	Results/Methods used	Research findings
Chan-drashekar et al. (2007) [316]	Biased and unbiased Hadamard walk	bit flip, phase flip, and generalized amplitude damping channels	Quantum information processing	line and cycle	one	Quantum trajectories approach	- symmetries preserved even in presence of noise - impact of noise models on walk dynamics - implementation on NMR(nuclear magnetic resonance) and ultracold atoms
Kumar et al. (2018) [248]	DTQW	uncorrelated static or dynamic disorder, spatial and temporal disorders	metrological domain, study dynamics of many-body systems	1-D lattice	one	- disorders are detrimental to quantum search - Von-Neumann entropy, coherence measure used for quantifying entanglement and interference of coin space respectively	- quantum memory enhanced using disorders - localization effects analyzed by monitoring information backflow - focuses on studies related to entanglement interference and non-Markovianity in quantum system
Colautti and Caruso (2019) [325]	DTQW	dephasing noise	energy transport in complex networks, quantum communication technologies	2-D graph, with four nodes	two	- Noise-assisted transport, and - transport efficiencies	- studies effects of noise on NAT, EAT and, - impact of initial entanglement on transport efficiency - EAT opens faster channels for slow walkers and, - EAT can avoid a broken link in a communication line.
Bruderer and Plenio (2016) [224]	CTQW, QSWM	Partial coherence ($\omega = 1/2$)	Graph isomorphism testing, modeling physical and biological processes	2-D graphs	single	- counting statistics of stochastic jumps - complex spectrum	- presented a quantum algorithm for GI testing resilient to decoherence - optimal performance at spectrum $\omega = 1/2$ ω spectrum provides additional information such as mixing time of QSW, algebraic connectivity of graph etc.
Wang et al. (2019) [321]	DTQW	Dynamic Quantum Phase Transitions(DQPT)	engineered non-unitary dynamics with decoherence, quench dynamics of many-body topological systems	1-D	single photon	Floquet operator U^f	- establishes direct relation between DQPT and DTOPs (dynamic topological order parameters) - investigates DQPTs for mixed quantum states and in-parity time-symmetric non-unitary quantum dynamics
Groh et al. (2016) [322]	DTQW	spin and spatial dephasing	energy transport in complex networks	1 and 2-D lattice	one	Floquet topological insulators	- developed a model for measuring robustness of topologically protected edge states with exponential decay. - experimental simulation to determine spatial limits between different topological phases.
Fedichkin and Meshchaninov (2016) [328]	CTQW	classic decoherence using quantum point contact observers	Quantum information processing	cycle	single electron	Relation obtained between mixing time, decoherence rate and graph size(n): $T_{mix,min} \approx 2/n, n \rightarrow \infty$ $T_{mix,min} \approx n \log(n), n \rightarrow \infty$, - decoherence rate for $T_{mix,min}$	- quantum to classical diffusion observed as decoherence rate increases - provides numerical and analytical expressions for dependency of decoherence rate and mixing time on the size of graph
Ghosal and Deb (2018) [326]	DTQW	dynamic percolation and phase-damping and bit-flip noise	Quantum search algorithms	square lattice	single walker	- For phase-damping noise: probability distribution of walker $\approx$ uniform distribution after infinite time - For bit-flip: probability distribution of walker varies greatly from uniform distribution even after infinite time	- results reconfirms that quantum walk spread faster than classical - quantum walk is faster if initial state is selected from Bloch sphere - closeness factor calculated for de-phasing noise and bit-flip, N=number of nodes in square lattice $\delta_{deph} = O(N^{-3/2})$ $\delta_{bit} = O(N^{-1})$

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Table 11 (continued).

Reference	Quantum walk type	Disorder type	Applications domain	Graphs used	N	Results/Methods used	Research findings
Geraldi et al. (2019) [311]	DTQW	space and time un-correlated tunable disorder	energy transport protocols in heterogeneous systems	1-D	two photons	- p is the value of disorder in the system - if $p > 0.40$, output distribution quickly merges into classical - for $0 < p \leq 0.20$, output distribution exhibits super-diffusive behavior	- provided experimental investigation of super-diffusive process, intermediate between classical and quantum regimes. - p-diluted quantum walk approach used to simulate decoherent process without changing the coherent evolution of particle. - sub-diffusive regime, between diffusive and localized regimes is also explored by increasing the number of step of quantum walk.
Zhan et al. (2017) [324]	non-unitary DTQW	periodical partial measurements on polarization of walker	exploring topological in-variance in physical systems	1-D	single photon	- average displacement of quantum walker is measured by exploring photon loss in walk evolution - detects two different invariants for non-unitary Floquet topological phases by using Floquet operators in distinct time-frames.	- first experimental investigation of bulk topological invariants for non-unitary discrete quantum walks. - confirms topological attributes of system by studying edge states at boundary of regions having discrete topological invariants.
Wang et al. (2018) [323]	DTQW photonic walk	chiral symmetry preserving disorder	explore topological phenomena with higher complex symmetries, quantum simulations	1-D	single walker	- measures chiral displacement and, - statistical moments of the walk dynamics	- detects winding numbers of walk using average chiral displacement - transitions in topological phases are identified - transitions confirmed by moments of particle probability distribution - observe localized edge states at region boundary's using winding numbers
Huang and Huang (2019) [327]	DTQW	calculated optimal disorder values	construct quantum networks robust against disorders and decoherence in quantum computing, biology, etc.	1 and 2-D	many particles	- particle-swarm optimization (PSO) method applied - results display 14% and 15% increase in average walking distance for one and two-dimensional quantum walk respectively	- applies PSO for obtaining optimal disorder for alleviating the effects of decoherence on walk dynamics - achieves substantial increase in mean walking distance for a broad range of decoherence parameter values of 0.01, 0.05 and 0.1
Chen and Zhang (2016) [320]	Alternative Quantum Walk	broken line noise and coin decoherence	quantum search algorithms	2-D	one 2-D walker	- decoherence effect appears only in one direction - transition from quantum to classical behavior is not obtained for all motions of quantum walker - classical mutual information and Measurement-induced Disturbance (MID) methods employed for correlation qualification	- anisotropic correlations between walker position and quantum coin state are revealed in two dimensional AQWs - analytic formulae for moments of position probability distribution of AQW are attained

N: Number of simultaneous quantum walkers in the system.

used applications include graph theory for doing similarity search on graphs [220–222] and metrology [204]. Noisy quantum walks on complex networks [230] and integrated photonic maze [231] have also been employed for experimental simulation of the quantum walk.

A detailed account of application domains and sub-domains along with their simulation results and summary has been provided in Table 12.

Citation count of an article can be seen as a measure of the usefulness, popularity and contribution in the related area. A graphical representation of the contribution of research papers based on citations and gradient-descent of citations in specific application domains is depicted in Fig. 15 where the vertical axis

denotes the percentage contribution of a particular article. Its corresponding table, i.e. Table 13 provides the details of the cited research articles.

5.2. Comparative analysis of random walk and quantum walk

In this section, we provide a comparative summary of the variation in computational cost and complexity for some well-known problems such as page ranking problem, graph isomorphism and others using two different approaches, the classical random walk and quantum walk approach. Table 14 provides a comparison between quantum walk and its classical counterpart for different existing problems in literature.

Table 12
Application domains of quantum walk.

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Hard problems	Graph collision problem	DTQW	–	Magniez et al. (2007) [167]	- proposed novice algorithm for finding triangle (K_3 copy) in graph G, having n vertices - design based on integration on quantum walk in Grover search	Query complexity	$\tilde{O}(n^{13/10})$	✗	–
Hard problems	Element distinctness problem	CTQW	general graphs	Ambainis (2007) [4]	- proposed quantum algorithm to find i equal items among n total items - generalized the query complexity to $O(n^{i/(i+1)})$	Query complexity	$O(n^{2/3})$	✗	single
Hard problems	Subset finding problem	DTQW	arbitrary graphs	Childs and Eisenberg (2003) [168]	- provided algorithm for finding a subset of size L in a graph with N nodes, based on [4] - several applications of proposed algorithm are presented such as element distinctness and finding L-clique in a graph.	Query complexity	$O(N^L/(L+1))$	✗	single
Hard problems	Hidden sub-graph problem	CTQW	three-ary tree	Zhang et al. (2018) [205]	- proposed a quantum based subgraph convolutional neural network (QS-CNNs) - it takes into consideration both global topological structure and also the local connectivity structure within a graph - used for graph and node classification tasks - used five datasets (MUTAG, PTC, NCI1, D & D, and PROTEINS)	Classification accuracy based on the parameters such as (i) learning rate, (ii) L2 regularization parameter, and (iii) drop-out rate	classification accuracy by 6.44% (WL), 9.89% (GK), 3.3% (PSCN-10 E) and 3.69% (FGSD) respectively, compared to the averaged classification of all competing methods over the five datasets	✗	–
Graph Isomorphism	–	DTQW	SRGs (strongly regular graphs)	Douglas and Wang (2008) [169]	devised DTQW based algorithms for identifying GI classes of large graph databases specifically strongly regular graphs	Complexity in terms of n (total nodes) and e (total edges) in graph G : (i) Time complexity (ii) State space complexity	(i) $TC = O(n^6)$, and (ii) $SSC = \log(2e)$	✗	single
Graph Isomorphism	–	CTQW	SRGs with up to 64 vertices	Gamble et al. (2010) [206]	- proved that interacting quantum particle worked better than non-interacting walkers in distinguishing pairs of non-isomorphic SRGs - provided analytical insight into the power of interacting two-particle quantum random walk by deriving distinguishing operators	δ , a measure of difference between matrix elements of evolution operator of pairs of graphs	(i) $\delta = 0$, isomorphic pair, and (ii) $\delta \neq 0$, graphs not isomorphic	✗	one and two
Graph Isomorphism	–	CTQW	SRGs	Rudinger et al. (2012) [207]	- explored the behavior of non-interacting multi-walker random bosonic and fermionic quantum walks in distinguishing isomorphic graph pairs in SRG family - Numerical simulations show that as the number of particles is increased the distinguishing power of quantum walks also increases, but it always remains inferior in comparison with interacting particle walk	success rate or distinguishing power based on boson and fermion failures	(i) total graph comparison = 70712, (ii) total failures = 256, and (iii) success rate $\geq 99.6\%$	✗	three
Graph Isomorphism	–	continuous QSWM	arbitrary simple, undirected graphs	Bruderer and Plenio (2016) [224]	- developed a quantum algorithm for graph isomorphism testing which provides optimal results when run in a partially coherent system as opposed to classical or fully coherent systems - QSW runs on entire graph and learns about the underlying topology, which is acquired by observing stochastic jumps across auxiliary edge	total jumps per graph is statistically analyzed to identify the ω spectrum of QSW	Isomorphism of two graphs G and G' is tested by comparing their spectra where different spectra imply non-isomorphic graphs	✓	single

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Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Graph Isomorphism	–	DTQW	regular graphs	Liu et al. (2019) [170]	- proposed an algorithm MapEff, for graph isomorphism detection problem via DTQW evolution on regular graphs - The algorithm shows improved accuracy of isomorphic mapping detection for regular graphs and uses auxiliary edges to distinguish symmetric nodes efficiently in polynomial time	(i) Computational complexity, and (ii) Accuracy of detection	(i) $O(n^6)$, n = total nodes in graph (G and G'), (ii) Accuracy ≥ 0.9 for regular graphs, and (iii) Accuracy = 1.0 for ordinary graphs	✗	single
Algorithms	Binary-NAND tree in Hamiltonian oracle	CTQW	binary NAND tree with N leaf nodes	Farhi et al. (2007) [208]	developed a quantum algorithm for evaluating binary NAND tree in Hamiltonian oracle model with N leaf nodes	Time complexity	$TC = \sqrt{N}$	✗	–
Meta-heuristic function	Simulated Annealing	DTQW	bipartite graphs	Somma et al. (2007) [171]	- proposed an algorithm for quantum simulation of classical annealing process, used to solve various Combinatorial optimization problems - The algorithm make use of principles of quantum walks, quantum Zeno effect and quantum-phase estimation methods	quantum speed-up in terms of minimum spectral gap (δ)	speed up of $1/\sqrt{\delta}$ in number of operations to find an optimal solution as compared to $1/\delta$ in classical approach	✗	–
Universal Computation	–	CTQW	low degree graphs(max degree= three)	Childs (2009) [117]	- provided a different approach for universal computation using a Hamiltonian acquired from adjacency matrix of a low-degree graph(3), against the earlier proposed time-independent Hamiltonian by Feynman [341] - The main logic is to implement quantum gates(CNOT, Hadamard gate, phase-shift, basis-changing graph widget, momentum filter, momentum separator) using scattering processes	the amplitude to propagate to vertex 0 on output line s	$ s\rangle U\rangle\langle 0 \times \Omega(1/m)$, for m -gate quantum circuit	✗	single
Universal Computation	–	CTQW	finite graphs	Childs et al. (2013) [209]	- A generic approach to more than one walker quantum systems proved that universal quantum computation is achievable using multi-particle systems with interactions on a polynomial size graph - The proposed approach can be used to design architecture for constructing scalable quantum computer without the requirement for time-dependent control	Error in simulating g -gate, n -qubit quantum circuit, wave-packet size to be polynomial function of n and g both	error is arbitrarily small constant, for (i) wave packet size = $O(n^{12}g^4)$, (ii) Total vertices = $O(n^{13}g^5)$, and (iii) Total evolution time = $O(n^{12}g^5)$	✗	multi
Universal Computation	–	DTQW	general undirected graph $G(V,E)$, maximum degree=8	Lovett et al. (2010) [118]	- presented DTQW version of earlier implemented quantum gates(CNOT, Hadamard gate, phase-shift, basis-changing graph widget, momentum filter, momentum separator) using CTQW by Childs [117] - It also confirms that both discrete and continuous-time models of quantum walks are computational primitives and thus computationally equivalent	Number of edges, wires and time steps for constructing quantum gates	(i) Edges(DTQW) = $2(\text{edges used in CTQW})$, (ii) Wires(DTQW) = Wires(CTQW), and (iii) Phase gate required one additional time step than in CTQW case	✗	single
Universal Computation	–	CTQW	1-D array	Peruzzo et al. (2010) [210]	- explored quantum correlations between two identical photons via CTQW evolution on linear array of vertices with fixed propagation and coupling coefficients - The observations reveal that the degree of correlation depends critically on the initial state of quantum walk	quantum correlations	classical limit violated by 76 standard deviations	✗	two

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Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Graph connectivity	connected components in graph	DTQW	directed reversible graphs	Montanaro (2005) [172]	• showed that reversibility of a directed graph is sufficient and necessary condition for allowing DTQW evolution and also • discussed some implications of the reversibility condition • provided an approach for partial quantum walk evolution on directed graphs that do not follow reversibility condition	Mathematical analysis	proof by lemma	✗	single
Graph connectivity	connected components in graph	DTQW	regular connected graphs	Albertini and Alessandro (2012) [173]	• devised a decentralized model of DTQW on a graph, in which the coin operator change with every vertex position of the graph at every time-step giving rise to control system • The results prove that controllability is verified only in case on non-bipartite graph, thus depending completely on the topological properties of the graph	Number of time steps to achieve controllability	atmost $(2k + r)$ steps, where k = number of coin operations, and r = order(conditional shift matrix)	✗	single
Graph connectivity	connected components in graph	Szegedy quantum walk	planar graph $G(n,e)$	Belows and Reichardt (2012) [225]	• used span programs to develop algorithms for problems such as st-connectivity, where either s and t are connected with a path of at-most length d or disconnected • developed algorithms to find whether a graph contains as sub-graph a star, path or sub-divided claw	Query Complexity(QC)	(i) $QC = O(n\sqrt{d})$, for st-connectivity, and (ii) $QC = O(n)$, for other algorithms, where n =total vertices in graph $G(n,e)$ and d = path length connecting vertices s and t	✗	-
Graph connectivity	detect marked vertices	Lackadaisical quantum walk	complete graph with self loop(k)	Wong (2015) [155]	• proposed a lackadaisical quantum walk by adding 'k' self loops to vertices of the complete graph • The effect of self loops on Grover's search for marked vertex on this complete graph with N vertices is made using discrete-time and continuous-time variants of quantum walk	for $G(N,e,k)$, N = total vertices, k = self loops (i) run time of algorithm, and (ii) success probability of finding marked vertex	for $N = 1024$, $k = 1, 2, 3$: (i) run time $\equiv 50, 41, 36$ (ii) success probability = 1, 0.889, 0.75	✗	single
Graph connectivity	detect marked vertices	DTQW	Jhonson graph	Magniez et al. (2011) [88]	• proposed a quantum search algorithm for finding marked elements in state space of Markov chain • It uses phase estimation algorithm to approximate reflection operator to be used for amplitude amplification	(i) Hitting time (H.T.), and (ii) Quantum Complexity in terms of setup cost(S), Update cost(U) and Checking cost(C)	(i) H.T = quadratic speed-up over classical algorithm (ii) Cost of order = $S + \frac{1}{\sqrt{\epsilon}}(\frac{1}{\delta}(U + C))$, ϵ =probability of finding a marked vertex and δ = eigenvalue gap of Markov chain	✗	single
Graph connectivity	detect marked vertices	DTQW	general graphs, 2-D grid	Krovi et al. (2010) [174]	• developed a novice approach for finding marked vertices in a graph using quantum walk evolution on graph with major operations, such as check, setup, update, shift etc. • The main concept applied is that of interpolation between random walk P and its absorbing walk P', with marked states as absorbing and quantum walk is quantum analogue of interpolation	Time steps to find marked vertex in M	$\sqrt{HT(P, M)}$, where HT =hitting time, p =quantum walk, and M =set of marker vertices	✗	single

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6. Discussion

This article presents a comprehensive survey of the recent research work done in the field of quantum walk and its application in related areas (as discussed in Sections 4 and 5). Our survey is supplementary to the other published surveys in this domain. Quantum walk evolution on graphs and its distinct models have been explored and reported using various taxonomies. Different coin operators used in discrete-time or coined quantum

walks have also been discussed. The article also provides in-depth knowledge about the experimental realization and impact of disorders and decoherence on quantum walk dynamics. Various existing software simulators and their running environment, programming language, limitations etc. have been explored and reported in this article. The major breakthroughs in the field of quantum walk started in the early twentieth century when algorithms based on quantum walks were designed and implemented

Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Graph connectivity	detect marked vertices	discrete ScQWM	complete graphs, multi-partite graphs	Hillery et al. (2010) [175]	- produced a novel application of scattering quantum walks, in finding a marked edge or marked complete subgraph of the given complete graph - produced a novel application of scattering quantum walks, in finding a marked edge or marked complete subgraph of the given complete graph	G(N,e,L), N = vertices in G, L = vertices in sub-graph of G;(i) steps to localize quantum walker in sub-graph (ii) search speed in number of oracle calls	(i) Steps = $O(N/L)$ (ii) quantum search quadratically faster than classical	✗	single
Graph connectivity	detect marked vertices	DTQW	general graphs	Tani (2008) [176]	- developed a quantum walk based algorithm that solves the claw finding problem optimally in terms of query complexity - The proposed algorithm can be generalized to find claw of n functions for any constant integer $n > 1$, where functions domain size may vary	Query complexity in number of queries to find claw in a graph	quadratic speed-up	✗	single
Graph connectivity	detect marked vertices	CTQW	Erdős-Rényi random graphs	Chakraborty et al. (2016) [211]	- developed an optimal spatial search CTQW based algorithm for finding a marked node in Erdős-Rényi random graphs, where graph has a total of n nodes and the probability for an edge to exist is p - quantum spatial search is proved to be optimal for almost all graphs where the ratio between the second largest and largest eigenvalue is bounded by a constant value < 1	Threshold value of p for optimal quantum search in $O(\sqrt{n})$ time	$p \geq \frac{1}{n} \log^{3/2}(n)$	✗	single
Artificial intelligence	human decision-making	quantum random walk	–	Sgarbas (2007) [227]	gives overview of recent advances in the field of quantum computation and its application to artificial intelligence	–	–	–	–
Graph connectivity	detect marked vertices	DTQW	general graph, non-vertex transitive structures, d-Cayley tree, Regular Hyper-branched Fractal(RHF)	Berry and Wang (2010) [177]	- studied the dynamics of DTQW to investigate the relation between success probability of finding a marked node in a graph and centrality of the node - further investigated relationship between frequency of success probability and marked node centrality - The frequencies present in success probability were proved to be associated with global structure of the graph and also the centrality and local structure surrounding the marked node	Time complexity of the search measured in terms of lowest frequency of success probability for 3-Cayley tree	T.C. = $O(N^{3.4}(\log N)^2)$, N = total number of vertices in graph	✗	single
Graph connectivity	detect marked vertices	StQWM	line graphs of bipartite, non-bipartite and bipartite graphs with perfect matching, and honeycomb lattice	Portugal (2016) [5]	- compared the three DTQW models, i.e. staggered walk model, Szegedy's model and coined models on special subclass of graphs using staggered formalism and search framework - Further, they provided the first implementation or example of a two-tessellable graph (iff its clique graph is bipartite) SQW search	Running time complexity for class 1 graph with 8-clique and 4-clique partial and complete tessellation respectively	Total running time with constant success probability, T.C = $O[N^{0.57}(\ln N)^{0.30}]$	✗	single

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using quantum systems. Later, quantum walk emerged as a universal computational framework, and many classical algorithms were implemented using quantum simulations.

In recent years, quantum walk framework of computation has gained momentum due to its experimental feasibility in physical systems. Quantum mechanics have revolutionized the fields of computation and information processing owing to its quantum

laws of nature which relate closely to physical, chemical and biological systems in nature such as photosynthesis process in plants [307]. The work on exploring properties of the quantum walk and their evolution on line, graphs, hyper-cubes and higher dimensional lattices, the effect of interacting and non-interacting quantum walkers on the walk dynamics and trade-off between quantum transport efficiency and disorders, and the presence

Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Graph connectivity	hidden non-linear structures	CTQW	Winnie Li graph	Childs et al. (2007) [212]	- provided a generalization of the well known Shor's factoring problem, namely to find hidden non-linear structures over finite field - Two instances of the above problem, i.e. hidden radius and hidden flats of centers problem are solved using quantum oracle and CTQW on Winnie Li graph respectively	Query Complexity	Quadratic speed-up over classical, $QC = poly(\log(q))$, where $q = p^m$, a prime power	✗	–
Search algorithms	spatial search	CTQW	complete graphs with broken links, bipartite and star graphs, binary tree and hypercube	Novo et al. (2015) [213]	- obtained a reduced set of states that encircles the dynamics of problem graph without knowledge of its underlying symmetries - applied invariant sub-space methods, which are computed systematically using Lanczos algorithm - The reduction method helped to obtain improved quantum transport efficiencies and an upper bound for accuracy of a single qubit transfer on XY spin network	Running time complexity of spatial search algorithm on a complete graph with broken links, and N vertices	$T.C = O(\frac{\pi(\sqrt{N})}{2})$	✗	single
Search algorithms	spatial search	DTQW	hypercube	Potoček et al. (2009) [178]	- presented improvement in the existing SKW (Shenvi, Kempe and Whaley's) quantum random walk search algorithm [3] for a finding a marked vertex using two alternative approaches - first approach finds the next neighbors of the target node with high probability - second approach deals with developing a two-coin quantum algorithm on a hypercube with self loops	(i) success probability (single marked vertex), (ii) query complexity, (iii) success probability for multiple marked vertices	(i) success probability ≈ 1 , (ii) $QC = \frac{1}{\sqrt{2}}(SKV)$ search, (iii) success probability $\approx \frac{1}{2}$	✗	single
Search algorithms	spatial search	CTQW	random temporal network, Erdős-Rényi random graphs	Chakraborty et al. (2017) [214]	- investigated the efficacy of quantum information tasks using CTQW on random temporal networks, i.e. networks whose topology changes in time - Given a random network on 'n' vertices and probability that an edge exists as p, after every time interval τ, the graph is replaced by a new graph $G(n,p)$ - algorithmic performance is non-trivially dependent on the interplay between connectivity and temporality of the graph	Running time complexity	$T.C = O(\sqrt{n})$, for $p_{threshold} = \frac{\log(n)}{\sqrt{n}}$, and a range of τ for $p_{threshold}$	✗	–
Search algorithms	unstructured or slightly structured problems	DTQW	arbitrary graphs	Childs and Eisenberg (2003) [168]	proposed a DTQW based subset finding problem in unstructured data set	Query complexity	$O(N^L/(L+1))$	✗	single
Search algorithms	unstructured or slightly structured problems	DTQW	general graphs	Tani (2008) [176]	developed a DTQW based algorithm that solves the claw finding problem optimally for unstructured data	Query complexity	quadratic speed-up over classical algorithm	✗	single
Graph partitioning	–	CTQW	multi-partite, and edge-weighted graphs	Carlson et al. (2006) [215]	- studied the dynamics of CTQW on edge-weighted graphs - derived results for instantaneous and average universal mixing and average almost-uniform mixing of quantum walks	Mixing times are calculated for: (i) weighted complete multi-partite graph, (ii) non-weighted complete graphs, and (iii) weighted graph with its spectrum not sub-linear in size of the graph	(i) instantaneous universal mixing, (ii) average universal mixing, and (iii) average almost-uniform mixing	✗	single

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Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Financial sector	Stock market prediction	quantum random walk	–	Araújo et al. (2008) [331]	proposed a quantum inspired intelligent hybrid (QIHH) method composed of ANN and modified quantum inspired evolutionary algorithm (mQIEA) for stock market forecasting	–	–	–	–
Cognitive science	human decision-making	CTQW (partially coherent quantum walks)	one-dimensional	Fuss and Navarro (2008) [216]	- introduced a class of partially-coherent quantum walk models for two-choice human decision making, used absorbing boundaries as psychological decision criteria - Studied effects of induced noise, such as phase and amplitude damping, and depolarization on quantum walk	Noise levels	(i) Noise = 0, pure quantum walk, (ii) Noise $\neq$ 0, Bernoulli random walk	✓	–
Cognitive science	human decision-making	CTQW	FMO complex (Fenna Matthews Olson)	Lloyd (2011) [217]	- light absorbed by photosynthetic bacteria creates exciton, which are electron-hole pair quantum systems, that follow quantum walk as they travel to reaction centers where this light energy can be converted to chemical energy - excitons employed a different type of quantum algorithm than quantum search, i.e. quantum walk for reaching reaction centers - They further studied the effect of surrounding temperature on the coherent quantum walk of exciton through FMO complex and proposed environmentally assisted quantum transport (ENAQT)	Effect of temperature on quantum walk dynamics in terms of (i) interference and (ii) quantum transport	(i) at low temperatures, disorders arise leading to destructive interference and Anderson localization, (ii) higher temperature resulted in more efficient transport and reduced destructive interference	✓	–
Cognitive science	human decision-making	quantum random walk	–	Biamonte et al. (2019) [226]	provides an insight into various application fields of quantum walks in complex networks, such as quantum transport, graph recognition and search engine ranking problems, for exciton transport in photosynthetic complexes	–	–	✗	–
Cognitive science	human decision-making	CTQW	1D lattice	Busmeyer et al. (2006) [218]	- developed a quantum equivalent of a Markov random walk model - Further, they developed its application to signal detection and obtained choice probability and response time for this model and also compared it to Markov model - The proposed quantum walk model can also be used to explain disjunction effects in the field of decision making	(i) Choice probabilities, and (ii) distribution of choice response time for a quantum walk model with parameters as mean = 2, Standard deviation = 1, phase = 20.5	(i) Final choice probabilities = 0.86, 0.14, and (ii) mean choice times = 38 ms, 28 ms	✗	single
Cognitive science	human decision-making	DTQW	line	Fuss and Navarro (2013) [179]	- Quantum model can be seen as a more natural interpretation in human decision-making owing to its cognitive architecture that is extremely parallel and involves both competitive (inhibitory) and co-operative (excitatory) interactions between units - presented the mechanistic psychological underpinnings of quantum walk model (CCP model (co-operative and competitive parallel) decision making model) applied to human decision-making and response time	Probability mass functions, and search optimization in terms of variance of model	(i) $\sigma_{CP}^2 \propto t$, CP = co-operative parallel model, t = number of time-steps, (ii) $\sigma_{CCP}^2 \propto t^2$, for CCP model	✓	single

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of decoherence in the system and its utilization to further improve walk metrics is expected to emerge as a focus area in the scientific research community in the coming years. An efficient, secure, scalable, physically realizable and cost-effective

quantum walk framework is required to address the growing demands in computationally sound future. Various open research questions and future perspectives in the field of quantum walk

Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Security	Stegano-graphy	DTQW	circle	EL-Latif et al. (2019) [180]	- proposed a novel technique to construct substitution boxes using DTQW with two walkers and two coin - also proposed an effective steganography method, which is an integration of classical data hiding and quantum walk evolution to provide improved security than existing state-of-the-art methods	performance criteria for quantum S-boxes are: (i) nonlinearity, (ii) bit independence (BIC), (iii) strict avalanche (SAC), (iv) BIC-SAC, (v) differential (DP), and (vi) linear (LP) approximation Probabilities	Performance criteria values: (i) nonlinearity = 106.25, (ii) BIC = 103.7, (iii) SAC = 0.5037, (iv) BIC-SAC = 0.5065, (v) DP = 0.0391, and (vi) LP = 0.1016	✗	two
Security	Stegano-graphy	DTQW	circle	Abd-El-Atty et al. (2019) [181]	- developed encryption and decryption protocols for NEQR (Novel Enhanced Quantum Representation) images using DTQW on a circle - A complete description of developed algorithms is provided using quantum circuits and digital computer simulations of common attacks on encrypted images is also presented - The proposed scheme has high efficiency and security along with a large key space	(i) Cardinality of key-space of t-step quantum walk, and (ii) Average key sensitivity	(i) $\sum_{t=t_1}^{t_n} 2^{t_n+1} - 2^{t_1}$, Cardinality $\equiv O(2^{t_n})$, (ii) 99.6090%	✗	single
Security	Stegano-graphy	DTQW	circle	Yang et al. (2015) [182]	- investigated nonlinear chaotic dynamics of QW in image encryption - constructed a novel highly secure test DTQW based algorithm for image encryption	(i) Speed analysis, (ii) Key space analysis, (iii) NIST SP800-22	(i) 0.1371721s for cipher images of size 256×256 , (ii) 2^{325} , (iii) pass	✗	two
Security	Stegano-graphy	DTQW	circle	Yang et al. (2016) [183]	- explored the capability of quantum hash function for secure key-distribution and pseudo-Random number generation (PRNG) by utilizing chaotic dynamics of quantum walks - proposed a novice image encryption quantum algorithm using quantum hash function	(i) mean changed bit number (B) (ii) probability (P), (iii) diffusion (δB) and (iv) confusion (δP), (v) Speed analysis, (vi) Key space, (vii) Correlation analysis	(i) B = 63.8137, (ii) $p \approx 50\%$, (iii) $\delta B = 5.6314$, (iv) $\delta P = 4.3931$, (v) 0.2 Mb/s, (vi) 2^{325} , (vii) correlation ≈ 0 (effective)	✗	single
Security	Stegano-graphy	DTQW	circle	Peng et al. (2019) [184]	- a novel image steganography technique is proposed to hide gray-scale secret image with a color cover image - quantum walk are applied for selection one channel of RGB in cover image to host secret bits	Mean values of: (i) PSNR (ii) SSIM (iii) UIQ (iv) NCC (v) capacity	(i) 48.5978 (ii) 0.9964 (iii) 0.9727 (iv) 0.996 (v) 2-bit/pixel	✗	–
Information Security	medical-image Steganogra-phy	Quasi-DTQWs	circle	Abd-El-Atty et al. (2020) [185]	- formulated a quasi-quantum-inspired approach where a secret image (color or gray-scale) is embedded onto a cover image (color or gray-scale) based on CAQWs for cloud e-healthcare - CAQWs are used to determine the position of pixels in carrier image to permeate the secret bits - proposed scheme does not require pre or post encryption of either secret or carrier images	(i) capacity/secret/ carrier) bits, (ii) Key space, (iii) statistical metric PSNR, SSIM, UIQ mean values	(i) capacity = (two/eight) bit, (ii) 10^{80} , (iii) 44.2314, 0.9499, 0.9487	✗	single

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and its applications have also been discussed in the subsequent sub-sections.

Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Security	IoT (Internet-of-Things)	DTQW, Controlled alternate quantum walks (CAQWs)	circle	Abd EL-Latif et al. (2019) [186]	- presented a new image (gray-scale and color) encryption mechanism (using CAQWs) for protecting patients privacy based on privacy preserving IoT based healthcare systems - Cryptosystem is composed of two parts, i.e. substitution and permutation implied using quantum walks	1. Information Entropy (i) Plain image, (ii) Cipher image 2. NPCR average value 3. Keyspace of encryption mechanism	1. IE (i) 7.1214 (ii) 7.9981, 2. 99.6197 3. 10^{160}	✗	two
Security	Security Protocol	DTQW	general graphs	Rohde et al. (2012) [187]	- proposed a protocol for client server communication and data processing (LOQC linear optics quantum computation) in boson sampling and quantum walk models - security of proposed protocol relies on the unique fact that there is no feed-forward or measurement within the circuit	Success probability	Prob. = $\sqrt{8/\pi m}$, m = number of photons in the system	✗	multi
Security	Crypto-graphy protocol	DTQW	line, circle	Vlachou et al. (2015) [188]	- proposed a highly secure protocol for quantum public-key cryptosystem - protocol uses entangled quantum states obtained from quantum walk evolution as public-keys	Efficiency of the proposed protocol	$\tau \approx \text{poly}(n)$, τ = overall time for protocol execution, n = length of the message	✗	single
Security	Security Protocol	DTQW	line	Chan-drashekar and Busch (2015) [189]	- proposed that a localized quantum walk with temporal disorder can be utilized to construct and retrieve or secure memory in a quantum system - quantum walk evolution can be utilized to retrieve the initial state of any unknown qubit in the system	probability of finding initial basis states $ 0\rangle$ and $ 1\rangle$, global phase factor Θ	$\Theta = \sum \theta_k \approx 0$	✓	single
Security	Security Protocol	DTQW	circle	Abd EL-Latif et al. (2019) [190]	proposed two efficient hash function methods(QWHF-1, QWHF-2) and security protocols, based on quantum walks for 5G networks which are highly secure, efficient and robust against many existing attacks	Statistical analysis: confusion and diffusion tests using mean change probability (P) and its standard variance $\delta(D)$	P = 50%, $\delta(D) \approx 7.1189$, for 1000 time steps	✗	single
Security	Hash function	DTQW	cycle graphs	Yang et al. (2019) [191]	- developed a generic quantum walk hash-function model (DHF), where the walker dynamics are controlled by two alternating coins - Swapping of coin operator is determined by a message bit at each time step	for n-bit message (i) Sensitivity of hash value w.r.t. message, (ii) Diffusion and confusion analysis	(i) for n = 33, hash value = 264 bits, high sensitivity, (ii) for one-bit change in original message, mean change of 132 bits and mean probability of change = 50% per bit, highly stable	✗	single
Teleportation	Communicating protocol	DTQW	line, circle with four nodes, complete graph with two vertices	Wang et al. (2017) [192]	- proposed a generalized quantum teleportation mechanism for realizing communicating protocol based on quantum walks with two coins - essential maximum entanglement source state appears during first step of walk and as such no prior entangled state is required	Number of measurements required to teleport a state having d-dimensions	two projective measurements required having d-elements	✗	single

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6.1. Open research questions and future perspectives

This section provides innovative research ideas and future directions to researchers aiming to work in this field. This

well-structured and authenticated framework would aid to build a strong base for further study and research in the field of quantum walk evolution and its applications. It would also be

Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Teleportation	–	DTQW	directed graphs	Chatterjee et al. (2019) [193]	- One, two and three qubit states teleportation implemented using a five qubit quantum computer and a 32-qubit simulator from IBM - the entanglement between coin and position space provides quantum channel for teleportation, i.e. transfer of state from one particle to other particle	mean probabilities (p) and standard deviation (σ) range for teleporting (i) $ 000\rangle$ and $ 111\rangle$ states, and (ii) $ 001\rangle$, $ 010\rangle$ and $ 100\rangle$ states	(i) $0 \leq p \leq 0.49$, $\sigma \approx 0.5$, (ii) $0 \leq p \leq 0.33$, $\sigma \approx 0.34$	✗	single
Teleportation	–	DTQW	line, circle, complete graph (with two nodes and loops)	Li et al. (2019) [194]	- developed a conclusive and flexible teleportation scheme of multiple arbitrary qubits using quantum walks - N-qubit arbitrary state can be teleported using only three graph structures - no requirement of prior entangled state preparation, it is done using multi-walker entanglement during walk evolution	deterministic probability for successful teleportation	$p = 100\%$	✗	multi (2, N)
Teleportation	Quantum Communication Protocols	DTQW	line, circle with N nodes, d-regular graph complete graph	Shang et al. (2019) [195]	- introduced perfect state transfer and teleportation methods using quantum walks with two coins - It was shown that information stored in coin1 state can be recovered perfectly at the target node only by atmost two times of coin flip operation on line and cycle graphs - Also, first perfect qubit transfer was implemented using quantum walk evolution with two coins on regular and complete graphs	Number of steps for teleportation to occur, success probability	steps = $2t$, $p = 1/2$, Unitary operation to recover transmitted state based on coin state	✗	single
Networking	Quantum Communication Protocols	DTQW	butterfly and inverted crown networks	Yang et al. (2018) [196]	- demonstrated a quantum multi uni-cast communication scheme over inverted crown network and butterfly networks - single qubit states are transmitted between multiple source sink pairs simultaneously with both probability and state fidelity value as one	(i) Success probability, (ii) State fidelity	(i) $p = 1$ (ii) fidelity = 1	✗	multi (two and three)
Artificial Intelligence	Machine Learning	QSWM	hypercube	Adcock et al. (2015) [228]	the study provides close association of quantum mechanics with machine learning and explores application domains, such as quantum neural networks and quantum stochastic walk	Quantum time scaling	No rigorous bounds	✓	–
Machine learning	Evolutionary algorithms	DTQW	binary-tree	Lovett et al. (2013) [197]	- proposed and compared algorithms based on differential evolution for adaptive many-particle quantum metrology and PSO (particle swarm optimization) and DE (differential equation) using AQEM (adaptive quantum enhanced metrology) parameter estimation - The proposed algorithm is then applied to binary decision tree model for enhanced interferometric phase estimation and to a decision tree for adaptive estimation of the unknown bias of a quantum coin in a quantum walk	for $d(\text{nodes}) = 4$, (i) Space cost, (ii) Time cost, (iii) Max number of walkers in PSO and DE without power-law breakdown, (iv) run-time improvement for using DE and not PSO	(i) $O(N)$, (ii) $O(N^7)$, (iii) $PSO = 35$, $DE = 75$, (iv) $(75/35)^7 \approx 208\text{fold}$ decrease in run-time	✗	multi

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Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Machine learning	Optimization	CTQW	–	Wossnig et al. (2018) [219]	- proposed a quantum algorithm QSVE (quantum singular value estimation) based on singular value estimation to find solution to linear systems for dense matrices - provided a solution vector x for equation $Ax = b$, A is a dense matrix and b is a vector	(i) Time complexity in run-time of algorithm, (ii) Run-time scaling of $n \times n$ dimensional A	(i) $TC = O(\sqrt{n})$, (ii) $O(\ A\ ^2 \sqrt{n} \text{polylog}(n)/\epsilon)$, $\ A\ $ = condition number of A , ϵ = precision parameter	✗	single
Graph theory	Similarity search	CTQW	graph datasets from bio-informatics	Bai et al. (2013) [220]	- showed the association of mixed quantum state with that of a graph and way to compute von-Neumann entropy of the corresponding density operator - the proposed quantum Jensen–Shannon graph kernel is defined as a function of the quantum Jensen–Shannon divergence between the graph density operators	(i) Time complexity, (ii) Accuracy analysis (a) MUTAG (b) ENZYMES (c) PPIs	$TC = O(V ^3)$ $ V $ is the total vertices in graph, (ii) Accuracy percentage: (a) 84.04% (b) 32.16% (c) 76.20%	✗	single
Graph Theory	Similarity search	CTQW	graph datasets from bio-informatics and computer-vision	Bai et al. (2015) [221]	- a novel quantum kernel is developed using Jensen–Shannon divergence technique (used for measuring the dissimilarity of quantum systems specified in terms of their density matrices) - Computation evaluation is done considering graph size and graph datasets size	Run-time for computing graph kernels with varying (i) vertex set size 'n' and (ii) size 'N' of graph datasets	(i) scales cubically with 'n' (ii) scales quadratically with 'N'	✗	single
Graph theory	Similarity search	CTQW	regular graphs	Rossi et al. (2015) [222]	- proposed a quantum algorithm to measure the similarity between a pair of unattributed graphs - two graphs are merged and the resultant graph is probed using continuous-time quantum walk using Jensen–Shannon divergence - divergence was maximum when the graphs were isomorphic and minimum when unrelated	(i) Computational complexity, (ii) Classification accuracy for (a) MUTAG and (b) SHOCK datasets	(i) $O(N V ^3)$, for dataset of N graphs with $ V $ vertices each, (ii)(a) Accuracy ≈ 88 (b) Accuracy ≈ 44	✗	two
Algorithm design	–	CTQW	sparse graphs	Low and Chuang (2019) [223]	- developed a quantum simulation algorithm to solve the problem of approximating the time-dependent evolution operator e^{-iHt} to error ϵ where Hamiltonian $H = (K\rangle \otimes \hat{U}(K\rangle \otimes \hat{I})$ is projection of unitary oracle $\hat{U}$ onto the state $K\rangle$, which is created by another oracle - the algorithm provides 'qubitization', in that it uses controlled version of the oracles to embed any Hamiltonian in an invariant $SU(2)$ space	(i) Query complexity, and (ii) Error probability	(i) $O(t + \log(1/\epsilon))$, (ii) $O(\epsilon)$ for 't' step and error ' ϵ '	✗	single
Metrology	–	DTQW	1-D lattice	Kitagawa et al. (2012) [204]	- simulation of one-dimensional topological phases using DTQW (split step quantum walk) - discovered a topologically protected pair of non-degenerate bound states	probability distributions of split-step quantum walks with and without decoherence	–	✓	single

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beneficial for the industry, as it provides new insights into quantum computer architecture and academia for research purpose. The open challenges and research question identified as per various quantum walk models and their applications in different domains (as mentioned in Sections 4 and 5) are discussed below.

6.1.1. Graph theory and hard problems

Quantum walk models have been widely used in formulating many hard problems such as graph collision, element distinctness problem, subset-finding and hidden sub-graph for which exponential time classical algorithms are available. Applying quantum walk to these problems significantly improves the computational power and provides a polynomial speed-up in time complexity.

Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Algorithm design	–	DTQW	line	Broome et al. (2010) [198]	- an intrinsically stable and deterministic implementation of discrete quantum walk is presented - quantum walk is measured up to 6 steps and also explores quantum to classical transition and also shows effects of absorbing boundaries taking into account decoherence	(i) Interference visibilities (IV), (ii) L_1 -norm distance for: (a) coherent walk (b) decoherent walk (iii) Optical loss(OL)	(i) IV $\approx$ 99.8% per step, (ii)(a) 0.031, (b) 0.017 (iii) OL $\approx$ 1% per step	✓	single
Information Processing	Quantum Neural Networks	QSWM	hypercube	Schuld et al. (2014) [229]	- constructed a stochastic quantum walk having associative memory between the global firing states of a QNN - speed-up in information transport in QNN, where a neuron is replaced by a qubit	Hamming distance	after n time-steps the quantum walk always finds sink closest to initial state in terms of Hamming distance	✓	single
Information Processing, Machine learning	Quantum Neural Networks	DTQW	general graphs	Dernbach et al. (2018) [199]	- proposed a novel quantum walk neural network on graphs - The model constructs a diffusion operator to be applied on graph-structured data for information flow throughout the graph - Experiments are conducted for node and graph regression and graph classification	(i) temperature prediction results for a walk of length 2, 3 and 4 (ii) Graph classification accuracy for datasets MUTAG, ENZYMES and NCI1 (iii) Atomization energy prediction for graph regression in terms of RMSE, MAE (iv) Space complexity	(i) $T \approx 5.54, 5.38, 5.28$, (ii) $QWNN_{centrality} \approx 0.32, 0.85, 0.68$ $QWNN_{similarity} \approx 0.32, 0.92, 0.71$, (iii) $QWNN_{centrality} \approx 12.52, 8.93$ $QWNN_{similarity} \approx 16.01, 10.78$ (iv) $\mathcal{O}(\mathcal{N}^2 d \mathcal{T})$, N is total nodes in graph, d is degree and T is the time steps of walk	✗	single
Information Processing	Genetic Algorithms	DTQW	hypercube	Pitchai et al. (2016) [200]	- problem of optimal team size is formulated as combinatorial optimization problem - a Quantum Walk based Genetic Algorithm (QWGA) is proposed to form near optimal teams that optimizes the fuzzy criteria obtained initially for team member skills	Maximum fitness values for 5 different fuzzified instances with total people as 100, 200, 500, 1000, 15000	Max. fitness = 24.6, 48.3, 98.7, 195.2, 220.9	✗	single
Information Processing	Genetic Algorithms	DTQW	line	Koppaka and Hota (2010) [201]	- proposed novel HQEA, i.e. quantum inspired evolutionary algorithms by introducing quantum Hadamard walk (QHW) for solving Combinatorial optimization problems such as 0/1 Knapsack - HQEA combines remote search, i.e mutation and local search	(i) RMS distance of classical and quantum for n=10 (ii) RMS distance of classical and quantum for n = 10 (iii) Average accuracy for size 200 and 500 for 0/1 Knapsack	(i) rms = 3.2(C), 5.5(Q) (ii) rms = 10(C), 54(Q) (iii) Accuracy = 1005.4, 2685.4	✗	single
Algorithm design	–	DTQW	regular graphs, Cayley graphs ergodic Markov chain	Szegedy (2004) [6]	- developed a generic method for quantizing classical algorithms such as Grover's search, element distinctness using diffusion based quantum walks - Prove that escape time depends on the spectral attributes of transition matrix with the marked rows and columns deleted	Average quantum hitting time and observation probabilities for $[n]^d$ torus, where n is total vertices and d is the dimension: (i) d = 1 (ii) d = 2 (iii) $d \geq 3$	(i) $n ; 1/n$ (ii) $n\sqrt{\log n} ; \frac{1}{\sqrt{\log n}}$ (iii) $n^{d/2} ; \Omega(1)$	✗	single
Machine learning	Forecasting	DTQW	line	Konno (2019) [202]	- proposed a novel non-stationary time series model based on quantum walks - oe-dimensional, two-state quantum walk dynamics are discussed	estimated values for x_2^* , given values $(x_0 = 0, x_1)$ for (i) $x_1 > 0$ (ii) $x_1 = 0$ (iii) $x_1 < 0$	x_2^* values: (i) $x_2^* = 1$ (ii) $x_2^* = x_1 = 0$ (iii) $x_2^* = -1$	✗	single

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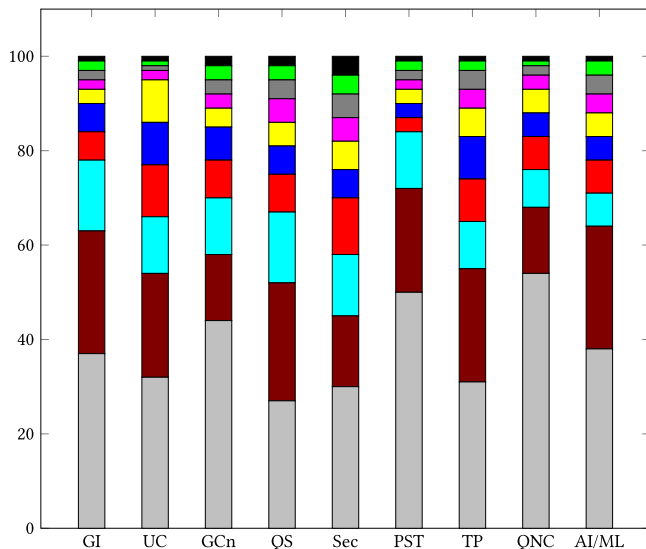
Table 12 (continued).

App domain	Sub-domain	QW model	Lattice type	Reference	Summary	Evaluation metric	Output	D	N
Quantum transport	Complex networks	QSWM	connected graphs without loops	Caruso (2014) [230]	- investigated the energy or information transfer (transport dynamics) between distant particles in noisy environment for large complex networks using Kossakowski-Lindblad formalism of quantum stochastic walks - studies transport properties of quantum stochastic walks	value of decoherence parameter ω for optimal transport efficiencies in linear (35 nodes), square grid (196 nodes), small (196 nodes) and large networks	$p \approx 0.1$, i.e. mixing ratio of 90% and 10% of quantum and classical dynamic respectively yield optimal transport efficiency	✓	single
Quantum transport	square lattice	QSWM	general graphs	Caruso et al. (2016) [231]	- demonstrates the way a walker reaches the output of a maze efficiently by partially suppressing the interference - optimal mixing of classical and quantum dynamics and controlled amount of decoherence results in efficient transmission of energy/information	optimal mixing ratio	$p \approx 0.1$, i.e. mixing ratio of 90% (quantum) 10% (classical)	✓	single
Security	Blockchain security	DTQW	circle	Chiang et al. (2020) [203]	- proposed a polyvalent, secure and client-centric quantum distributed ledger system using an asymmetric Multiple Attribute Lock Encryption (MALE), which is further used for designing key-chain update protocol - quantum walks are simulated to generate keys with high randomness quotient using pseudo random number generation mechanism - Quantum Key Distribution (QKD) protocol is employed to serve as seed provider and avoid problems such as double spending	probability to guess secret-key by an outsider	$p = \max(\frac{1}{c(4T, T)}, \frac{1}{2^T})$, T is bits generated from BB84 algorithm	✗	single

App Domain: Application Domain.

QW Model: Type of quantum walk model.

D: Decoherence.

**Fig. 15.** Research contribution of articles in different domains. GI: Graph Isomorphism, UC: Universal Computation, GCn: Graph Connectivity, QS: Quantum Search, Sec: Security, ML: Machine Learning, PST: Perfect State Transfer, TP: Teleportation, AI: Artificial Intelligence, QNC: Quantum Communication Protocols.**Table 13**

Contribution table representing different articles based on citation count. AD: Application Domain, Col: Color, GI: Graph Isomorphism, UC: Universal Computation, GCn: Graph Connectivity, QS: Quantum Search, Sec: Security, ML: Machine Learning, PST: Perfect State Transfer, TP: Teleportation, AI: Artificial Intelligence, QNC: Quantum Communication Protocols [342–389].

Col	AD								
	GI	UC	GCn	QS	Sec	PST	TP	QNC	AI/ML
	[210]	[342]	[2]	[343]	[182]	[344]	[345]	[210]	[346]
	[17]	[117]	[347]	[241]	[187]	[348]	[192]	[310]	[349]
	[283]	[350]	[351]	[88]	[352]	[353]	[288]	[354]	[355]
	[356]	[118]	[36]	[357]	[358]	[359]	[195]	[360]	[361]
	[206]	[362]	[7]	[343]	[181]	[363]	[364]	[59]	[128]
	[365]	[209]	[92]	[178]	[366]	[367]	[368]	[369]	[370]
	[111]	[371]	[372]	[47]	[373]	[374]	[375]	[187]	[376]
	[207]	[107]	[172]	[211]	[377]	[378]	[379]	[249]	[200]
	[390]	[380]	[381]	[177]	[247]	[382]	[194]	[296]	[199]
	[383]	[384]	[5]	[385]	[386]	[387]	[388]	[82]	[389]

But, application of quantum walk models may prove to be less cost-efficient and as such a trade-off between the resources and computation time needs to be established. Also, most of the algorithms are designed using a discrete-time model, although continuous-time formulation may represent the problem more naturally and even gets rid of the additional coin register. Most of the existing literature focuses on quantum walk evolution on simple undirected graphs. Thus, reducing the existing problems into more suitable graph structures whether regular or irregular,

Table 14
Comparison between random walk and quantum walk based on the type of application.

Problem definition	Evaluation metric	Random walk	Quantum walk	Reference
Page ranking	Time Complexity	$\mathcal{O}(\frac{\sqrt{\log N}}{\epsilon})$ N = network size ϵ = reset probability	faster than classical page rank algorithm, exact values for networks of various sizes is provided for different networks (scale free, outerplanar hierarchical)	Sarma et al. [391] Chawla et al. (2020) [392] Loke et al. (2017) [393]
Searching for marked vertices	Time complexity	TC is polynomial in number of vertices in the graph	Quadratic speed-up over classical case	Santha [16]
Graph Isomorphism	(i) Time complexity (ii) Space complexity C = constant, N = number of vertices in G, e = edges in graph G	(i) TC = $\mathcal{O}(C^{\sqrt{N} \log N})$ (ii) SC = $\mathcal{O}(n^2)$	(i) TC = $\mathcal{O}(n^5 \log(n)^2)$ (ii) SC = $\log(2e)$ qubits	Douglas and Wang (2008) [169]
Graph Propagation G_n	Propagation time between two nodes of G_n , n is total nodes in G	$\mathcal{O}(e^n)$	$\mathcal{O}(n)$	Childs et al. (2002) [394]
Simulated Annealing	(i) Resources complexity $\mathcal{N}$ (ii) Number of operations δ is spectral gap ϵ is error probability d = $\mathcal{O}(\exp(N))$, is state space for graph with N vertices	(i) $\mathcal{N}_{SA} = \mathcal{O}(\log(\frac{d^2}{\epsilon})/\delta)$ (ii) $\frac{1}{\delta}$ operations	(i) $\mathcal{N}_{QSA} = \mathcal{O}(\frac{\log^3(2d/\epsilon^2)}{\epsilon^2 \sqrt{\delta}})$ for $\delta \ll 1$ (ii) $\frac{1}{\sqrt{\delta}}$ operations	Somma et al. (2007) [171]
Element Distinctness	Query complexity for k equal numbers in N-vertex graph	–	$\mathcal{O}(N^{\frac{k}{k+1}})$	Ambainis (2007) [4]
Bipartite matching	Accuracy rate for COIL-100 data set	Accuracy $\geq 92\%$	Accuracy $\geq 99\%$ (CTQWM) and $\approx 100\%$ (DTQWM) and faster execution with lower computation cost	Gori et al. (2005) [395] Emms et al. (2009) [390,396]
Decision trees traversal	Traversal time	exponential in n n = number of vertices in tree T_n	$\mathcal{O}(n)$	Farhi and Gutmann (1998) [97]

directed or undirected and with or without loops and graph categorization based on heat kernels and commute time is another future direction of research. Problems such as st-connectivity or vertex reachability problem have been efficiently solved using a quantum walk, but studying the impact of multiple walkers on the speed-up in hitting time and the impact of, node ordering, disorders and dynamically changing graph topology on the time complexity of the model can be seen as a critical area for research. Thus, there is a need to explore these aspects in studying quantum walk dynamics on graphs.

6.1.2. Graph isomorphism

Graph isomorphism testing and similarity search have been one of the most explored domain in classical random walk and quantum walks. Finding the optimal test suite for checking of graph isomorphism and testing of chiral and regular graphs remains an open area of research. Testing of graph pairs with dynamically changing topology and with missing edges or links can be another research domain.

6.1.3. Design of quantum algorithms

Development of algorithms based on quantum walks and quantum simulation of the classical algorithm has been widely studied by the research community. As such developing algorithms for simulating naturally occurring processes such as food foraging behavior of the insects, photosynthesis in the plant, chemical reactions using quantum walk formulation can be an exciting area of research in this field.

6.1.4. Universal computation

Quantum walk has emerged as a universal computation model in recent years. Experimental implementation of quantum walks in physical systems using optical lattice, trapped ions, interferometer has also been widely explored. Using quantum walk formulations for realizing quantum gates such CNOT, Hadamard

gate and quantum circuits have gained popularity recently with its likely-hood in designing quantum computers which makes it essential areas of future research.

6.1.5. Security

Security is a critical parameter in various application domains such as crypto-currency, block-chain, cloud management, Internet of things framework, e-Healthcare, financial sector and communication. Although, some of these domains have been studied from quantum walk perspectives such as designing encryption algorithms, steganography techniques and key management methods, there are still many research dimensions that are left unexplored in these domains including bitcoin security, designing efficient communication protocols, block-chain security, among others. Thus, this sub-domain must be explored to provide effective security mechanisms in various areas.

6.1.6. Machine learning

Exponential increase in the size of data has made the machine learning domain as the most researched domain. Searching the data space for finding useful information forms the core of such algorithms. Quantum walk models have proved to provide quadratic speed-ups in detecting and finding marked data nodes. Thus, developing hybrid algorithms using existing machine learning approaches such as genetic algorithms and other evolutionary algorithms together with quantum walk evolution can expand the application domain of both the domains and provide promising results. This field can thus be regarded as a potential research direction in the industry as well as academia.

6.1.7. Transition between different quantum walk formulations

Different quantum walk models have been studied in literature such as discrete-time, continuous-time, stochastic, and staggered. Two major model classifications are discrete-time which operates in Hilbert coin and position space using coin and walkers position respectively, and continuous-time wherein quantum

walk evolves using as per time-dependent Hamiltonian. Transitioning from discrete to continuous eliminates the need for up-keeping the coin register, thus improving the space complexity substantially for larger networks. Since quantum walks can be seen as a universal model of computation, so the need for establishing mathematical and theoretical proofs for transitioning between these models has become an essential part of this field. Studying the prospective environment and equation for the feasibility of this transition and the constraints has become inevitable and this can be seen as an upcoming research direction in the near future.

6.1.8. Impact of multiple walkers and coins

Majority of the existing research work focuses on evolution of a single quantum walker in a quantum system. Experimental results have proved that using more than one walker may significantly improve the hitting time in some applications whereas, there is no impact on the spreading behavior of quantum walk in some cases. Exploring different application domains to study the impact of multiple walkers, whether independent or interacting and their correlation on the quantum walk dynamics can be seen as potential research dimension. Additionally, the use of different coin operators in an alternating manner, such as in the case of controlled alternate quantum walks has been widely applied in spatial search algorithms. But, exploring the impact of multiple coin operators in walk evolution and their applicability constraints in terms of space and complexity of implementation can also be addressed as an important research topic in this field.

6.1.9. Effect of decoherence on walk dynamics

Decoherence and presence of partial disorders in quantum systems have proved to improve the transport efficiency in quantum walks and resulted in perfect state transfer of qubit states. However, seeding decoherence in the quantum system and exploring the optimal value of the parameter for efficient transport can be seen as a critical area of research. Determining the coefficient of decoherence in various application domains as quantum transport in complex networks and its relation to classical-quantum transition may open new insights in the applicability of quantum walk models in more naturally occurring phenomena such as meteorology, many-body systems and other biological processes.

6.1.10. Quantum walks and optimization

Quantum walks have emerged to be at the core of innovative quantum technologies. Quantum walks have been recently introduced in the area of optimization problems such as minimization problems, discrete optimization problems and other constraint satisfaction problems. Furthermore, the relation between quantum walks and optimization problems such as quantum speed-up problem and optimal adiabatic schedule calculation for hypercubes by CTQW have also been investigated lately. Although some appreciable research has been carried out in this domain, still many aspects such as various optimization functions along with their advantages and limitations, different categories of graphs suitable for optimization problem formulation, *etc.* need more exploration. As such, quantum walk application in the field of optimization problems calls for further research and analysis in this domain.

7. Conclusion

In this article we have systematically reviewed the theoretical and practical aspects of various quantum walk models. A firm mathematical foundation and importance of various performance metrics are also discussed for quantum walk evolution on a line

and graphs. A brief summary of the papers published on different quantum walk models such as discrete-time, continuous-time, open, lazy, quantum stochastic walks and others is also presented. The paper also highlights the widely used coin operators in discrete-time model along with their complexity, strengths, mathematical formulation and application domains. Various existing simulators for implementing quantum walk models and their working environment description with limitations are also discussed in the paper. Moreover, we have reviewed the experimental proposals and implementation of quantum walks in physical systems. The effect of decoherence and noise on quantum walk dynamics has also been studied and reported. Furthermore, we have extensively investigated the innovative application areas of different quantum walk models and summarized their contribution and results for in-depth understanding of the field. Research gaps and challenges in application and physical implementation of quantum walks have been identified and reported in this survey. Fortunately, quantum walk has emerged as a core field of research in computational and information processing and is full of open challenges for aspiring researchers from many domains such as physics, computer science, engineering and biology among others. This survey, which is meant to be considered as a contribution to the field of quantum walk and their applications from a computer science perspective is an addition to the existing similar reviews with a focus on recent developments and findings in this field. Lastly, future research directions in this discipline have also been discussed. The article will best serve the research community if it encourages aspiring quantum scientists, researchers, people from industry and academia to further advance on this research discipline.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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